United States Environmental Protection Agency

EPA/600/R-23/145 | December 2023 | www.epa.gov/research

PM_{10} Supplement to the 2021 Report on Performance Testing Protocols, Metrics, and Target Values for Fine Particulate Matter Air Sensors

Office of Research and Development
Center for Environmental Measurement and Modeling

EPA/600/R-23/145 | December 2023 | www.epa.gov/research

PM_{10} Supplement to the 2021 Report on Performance Testing Protocols, Metrics, and Target Values for Fine Particulate Matter Air Sensors

By

Rachelle Duvall, Andrea Clements, and Karoline Barkjohn
Center for Environmental Measurement and Modeling
Office of Research and Development
U.S. Environmental Protection Agency
Research Triangle Park, NC 27711

Menaka Kumar
National Student Services Contract
U.S. Environmental Protection Agency
Research Triangle Park, NC 27711

Danny Greene
Eastern Research Group, Inc.
Morrisville, NC 27560

Tim Dye
TD Environmental Services, LLC
Petaluma, CA 94952

Vasileios Papapostolou, Wilton Mui, and Michelle Kuang
South Coast Air Quality Management District
Diamond Bar, CA 91765

Disclaimer

This report presents work performed by the U.S. Environmental Protection Agency's (U.S. EPA) Office of Research and Development (ORD) with technical support provided by the Eastern Research Group through two task orders: Task Order 68HERH19F0257 and Task Order 68HERH21F0093 both under EPA Contract No. 68HERD19A0001. This effort represents a collaboration between ORD, the U.S. EPA Office of Air Quality Planning and Standards (OAQPS), and the South Coast Air Quality Management District (South Coast AQMD) Air Quality Sensor Performance Evaluation Center (AQ-SPEC). Any mention of trade names, manufacturers or products does not imply an endorsement by the U.S. Government or the U.S. EPA. The U.S. EPA and its employees do not endorse any commercial products, services, or enterprises. This document has been reviewed in accordance with U.S. EPA policy and approved for publication. Testing results do not constitute certification or endorsement by the U.S. EPA.

Table of Contents

Disclaimer iii

Table of Contents iv

List of Figures vi

List of Tables vi

Acronyms and Abbreviations viii

Acknowledgements x

Executive Summary xi

1.0 Introduction 1

1.1 Background 1

1.2 Motivation 2

1.3 Objectives 2

2.0 Performance Testing Protocols for PM_{10} Air Sensors 4

2.1 Base Testing 4

2.1.1 Materials and Equipment 7

2.1.2 Selecting and Setting up a Test Site 8

2.1.3 Setting up the Air Sensors 10

2.1.4 Conduct Base Testing 10

2.2 Enhanced Testing 10

2.2.1 Materials and Equipment 13

2.2.2 Equipment Set Up in Exposure Chamber 15

2.2.3 Initial Testing Conditions 15

2.2.4 Effect of Relative Humidity (RH) 15

2.2.5 Effect of Temperature (T) 16

2.2.6 Drift 16

2.2.6.1 Drift (Day 1) – Low and Mid Concentration Drift 16

2.2.6.2 Drift (Day 60) to Evaluate Sensor Aging 17

2.2.7 Accuracy at High Concentrations 18

3.0 Performance Metrics and Supporting Calculations for Evaluating PM_{10} Air Sensors 19

3.1 Base Testing Calculations 20

3.1.1 Daily Averages 20

3.1.2 Deployment Averages 20

3.1.3 Precision 20

3.1.4 Bias and Linearity 20
3.1.5 Error 20
3.1.6 Exploring Effect of Meteorology 20
3.1.6.1 Potential Scatter Plots 20
3.1.6.2 Normalized Concentration 21
3.1.6.3 Concentration Difference and Absolute Concentration Difference 21
3.1.6.4 Dew Point (DP) 21
3.1.7 $PM_{2.5}/PM_{10}$ Ratio 21
3.2 Enhanced Testing Calculations 21
3.2.1 Data Averages 22
3.2.2 Test Averages 22
3.2.3 Precision 22
3.2.4 Bias and Linearity 22
3.2.5 Error 22
3.2.6 Effect of Relative Humidity (RH) 22
3.2.7 Effect of Temperature (T) 22
3.2.8 Drift 22
3.2.9 Accuracy at High Concentrations 22
3.2.10 $PM_{2.5}/PM_{10}$ Ratio 22
4.0 Target Values for PM_{10} Air Sensors 24
4.1 Approach 24
4.2 List of Target Values 25
5.0 References 28
Appendix A: Definitions 30
Appendix B: Supporting Information for Testing Protocols 31
Appendix C: Supporting Information for Performance Metrics 40
Appendix D: Supporting Information for Target Values 41
Appendix E: Checklist for Base Testing 52
Appendix F: Example Reporting Template for Base Testing 53
Appendix G: Checklist for Enhanced Testing 56
Appendix H: Example Reporting Template for Enhanced Testing 57

List of Figures

Figure 2-1. Example Approach for Conducting Base and Enhanced Testing of a Single Set of Air Sensors 4

Figure 2-2. Overview of the Enhanced Testing Procedures for PM_{10} Sensors 11

Figure B-1. Average PM_{10} Concentrations for Select Months over 2017-2019 across the Continental U.S. 32

List of Tables

Table ES-1. Summary of the Similarities and Changes for Base (B) and Enhanced (E) Testing of $PM_{2.5}$ and PM_{10} Air Sensors (gray shaded cells indicate changes) xiii

Table ES-2. Base and Enhanced Testing – Recommend Performance Metrics and Target Values for $PM_{2.5}$ and PM_{10} Air Sensors xiv

Table ES-3. Enhanced Testing – Additional Recommended Performance Metrics and Test Conditions for $PM_{2.5}$ and PM_{10} Air Sensors xv

Table 2-1. Summary of the Key Similarities and Differences for Base Testing of $PM_{2.5}$ and PM_{10} Air Sensors (gray boxes indicate differences) 7

Table 2-2. Test Site Selection Criteria for PM_{10} Sensors 9

Table 2-3. U.S. Regional PM_{10} Trends from 2018-2022 (Source: National Air Quality Status and Trends of Key Air Pollutants; available at *https://www.epa.gov/air-trends*, last accessed 05/30/2023) 9

Table 2-4. Summary of the Key Similarities and Differences for Enhanced Testing of $PM_{2.5}$ and PM_{10} Air Sensors (*gray shaded boxes indicate differences*) 13

Table 2-5. Initial Testing Conditions for PM_{10} Sensors 15

Table 2-6. Elevated RH Test Conditions for PM_{10} Sensors 16

Table 2-7. Elevated T Test Conditions for PM_{10} Sensors 16

Table 2-8. Low Concentration Drift Test Conditions for PM_{10} Sensors 17

Table 2-9. Mid Concentration Drift Test Conditions for PM_{10} Sensors 17

Table 2-10. High and Higher PM_{10} Concentration Test Conditions 18

Table 3-1. Summary of Recommended Performance Metrics for $PM_{2.5}$ and PM_{10} Sensors (*gray boxes indicate differences*) 19

Table 4-1. PM_{10} Sensor Performance Field Evaluation Results from Available Resources 25

Table 4-2. Base and Enhanced Testing – Recommended Performance Metrics and Target Values for $PM_{2.5}$ and PM_{10} Sensors Used in Ambient, Outdoor, Fixed Site NSIM Applications 26

Table 4-3. Enhanced Testing – Additional Recommended Performance Metrics and Test Conditions for $PM_{2.5}$ and PM_{10} Sensors Used in Ambient, Outdoor, Fixed Site NSIM Applications 27

Table B-1. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Winter Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} concentrations are ≥ 40 µg/m^3 at least once. 34

Table B-2. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Spring Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} concentrations are ≥ 40 µg/m^3 at least once. 35

Table B-3. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Summer Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} Concentrations are ≥ 40 µg/m^3 at least once. 36

Table B-4. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Fall Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} Concentrations are ≥ 40 µg/m^3 at least once. 37

Table D-1. Performance Requirements for PM_{10} FRM/FEM Regulatory Monitors 41

Table D-2. Summary of U.S. EPA PM_{10} Sensor Evaluations 42

Table D-3. Summary of AQ-SPEC PM_{10} Sensor Field Evaluation Results (1-hour Average) 43

Table D-4. Summary of AQ-SPEC PM_{10} Sensor Field Evaluation Results (24-hour Average) 45

Table D-5. Summary of Literature Reviews of PM_{10} Sensor Evaluations (1-hour average) 47

Table D-6. Summary of Resources Used to Inform PM_{10} Target Values 48

Acronyms and Abbreviations

AMTIC	Ambient Monitoring Technology Information Center
AQI	Air Quality Index
AQMD	Air Quality Management District
AQ-SPEC	Air Quality Sensor Performance Evaluation Center
AQS	Air Quality System
b	intercept
CEMM	Center for Environmental Measurement and Modeling
CFR	Code of Federal Regulations
CV	coefficient of variation
°C	degrees Celsius
DL	detection limit
DP	dew point
Eq	equation
FEM	Federal Equivalent Method
FRM	Federal Reference Method
IMPROVE	Interagency Monitoring of Protected Visual Environments
ISO	International Organization for Standardization
m	slope
MEE	Ministry of Ecology and Environment (People's Republic of China)
NAAQS	National Ambient Air Quality Standards
NAPS	National Air Pollution Surveillance (Canada)
NCore	National Core Network
NIST	National Institute of Standards and Technology
NRMSE	normalized root mean square error
NSIM	non-regulatory supplemental and informational monitoring
OAQPS	Office of Air Quality Planning and Standards
ORD	Office of Research and Development
PM	particulate matter
$PM_{10-2.5}$	coarse particulate matter, also abbreviated as PMc; inhalable particles with diameters generally larger than 2.5 micrometers and smaller than, or equal to,10 micrometers in diameter
$PM_{2.5}$	fine particulate matter, also abbreviated as fine PM; inhalable particles with diameters that are generally less than 2.5 micrometers
PM_{10}	inhalable particles with diameters that are generally less than 10 micrometers
QA	quality assurance
QAPP	Quality Assurance Project Plan
QC	quality control
r	Pearson correlation coefficient
R^2	coefficient of determination
RH	relative humidity
RMSE	root mean square error
SD	standard deviation
SLAMS	State and Local Air Monitoring Station
T	temperature
µm	micrometers

$\mu g/m^3$	micrograms per cubic meter
U.S.	United States
U.S. EPA	United States Environmental Protection Agency

Acknowledgements

The authors acknowledge the Eastern Research Group technical staff associated with Task Order 68HERH19F0257 and Task Order 68HERH21F0093 (both under EPA Contract No. 68HERD19A0001) for their research efforts summarized here. This effort was led by the United States Environmental Protection Agency (U.S. EPA), Office of Research and Development (ORD), Center for Environmental Measurement and Modeling (CEMM) with support from the Air Sensor Group that consisted of staff in ORD/CEMM. We acknowledge the Subject Matter Experts that participated in the U.S. EPA's Air Sensors 2019 second workshop on "Deliberating Performance Targets for Air Quality Sensors", whose feedback helped inform this report. We acknowledge Samuel Frederick (former National Student Services Contractor assigned to U.S. EPA/ORD/CEMM), Ashley Collier-Oxandale (formerly with South Coast AQMD), and Brandon Feenstra (South Coast AQMD) for contributing to data analysis used in this report. Libby Nessley (U.S. EPA/ORD/CEMM Quality Assurance) is recognized for quality assurance support in developing this report.

Lastly, we acknowledge U.S. EPA internal reviewers Robert Vanderpool and Tim Hanley. We also acknowledge the following external reviewers: Janice Lam Snyder (Sacramento Metropolitan Air Quality Management District) and Paul Fransioli (Clark County Nevada, Department of Environment and Sustainability).

Title page photo by South Coast AQMD as part of work under the U.S. EPA Contract #EP-16W-00017. Photo shows the custom-built Citizen Science Air Monitors at the Rubidoux Air Monitoring site in Riverside, California.

Executive Summary

Air sensors continue to be more accessible nationwide, and their development is evolving at a rapid pace. The use of air sensors is more widespread for a variety of air monitoring applications and data sets are readily available to the public. While continued innovations in air sensors make them more useful for air quality monitoring, the data quality remains highly variable. In response to the widespread use of air sensor technologies, variability in sensor data quality, and the lack of consistent testing protocols to evaluate sensor performance, the United States Environmental Protection Agency (U.S. EPA) published a report in 2021 outlining recommended testing protocols to evaluate the performance of fine particulate matter ($PM_{2.5}$) air sensors for non-regulatory supplemental and informational monitoring applications (NSIM; see Section 1.1) in ambient, outdoor, fixed site environments (U.S. EPA, 2021). In addition to the testing protocols, this report includes recommended performance metrics, instructions on how to calculate the metrics, and corresponding target values (testable performance metrics) to understand and compare sensor performance. The report also contains reporting templates to facilitate consistent communication of performance evaluation results.

The intended audience for the 2021 report and this supplement is the same and includes potential testing organizations, sensor manufacturers, and sensor developers. It is anticipated that a variety of consumers (e.g., state/local/Tribal agencies, federal government agencies, community groups, participatory/citizen scientists, academia) will benefit from the consistent presentation of testing results to identify sensor technologies that would be best suited for their NSIM application and understand the performance of the air sensor technologies. Consumers may also choose to conduct these testing protocols.

Initially, guidance was provided for $PM_{2.5}$ sensors as there was more widespread use and more technical information and knowledge available for these technologies. Often many sensors that measure $PM_{2.5}$ also report concentrations for inhalable particles with diameters that are generally less than 10 micrometers (PM_{10}). However, the ability of sensors to reliably measure PM_{10} is more uncertain. Known measurement issues in existing PM_{10} sensors are related to poor aspiration of coarse particles and efficiency of optical detection methods. Measurements can also be impacted by temperature and relative humidity. Fewer performance evaluations and successful field applications have been reported by testing organizations and in scientific literature for PM_{10} sensors suggesting a need for further development, testing, and improvements. Despite these issues, the U.S. EPA recognizes that consumers are interested in using PM_{10} sensors for a variety of NSIM applications. Additionally, testing protocols, performance metrics, and target values can be useful for evaluating the performance of existing PM_{10} sensors and helping guide future technology improvements.

The objective of this supplemental report is to build on the guidance provided in the 2021 report titled “Performance Testing Protocols, Metrics, and Target Values for Fine Particulate Matter Air Sensors”, herein referred to as the “$PM_{2.5}$ Targets Report”. This PM_{10} supplement provides updates to the testing protocols, metrics, and target values to evaluate the performance of PM_{10} sensors for NSIM applications in ambient, outdoor, fixed site environments.

Guidance on evaluating PM_{10} sensors is provided as a supplement as there is significant overlap with the $PM_{2.5}$ Targets Report. This format is intended to highlight new information, as applicable, and the key differences and considerations for evaluating PM_{10} sensors. Similar to the $PM_{2.5}$ Targets Report, base testing (field) and enhanced testing (laboratory) are recommended for PM_{10} sensors. A comparison of the changes for evaluating PM_{10} air sensors is shown in Table ES-1 with additional details in Section 2.0.

Table ES-1. Summary of the Similarities and Changes for Base (B) and Enhanced (E) Testing of $PM_{2.5}$ and PM_{10} Air Sensors (gray shaded cells indicate changes)

Category (Test Type)		$PM_{2.5}$	PM_{10}	Comments
Equipment (B, E)	Sensors	Three (3) sensors of the same make, model, and firmware version	Three (3) sensors of the same make, model, and firmware version	No change in recommendations
	FRM/FEM monitors	Calibrated $PM_{2.5}$ FRM/FEM monitor	Calibrated PM_{10} FRM/FEM monitor Calibrated $PM_{2.5}$ FRM/FEM monitor	Both a calibrated $PM_{2.5}$ and PM_{10} FRM/FEM monitor are suggested to calculate a $PM_{2.5}/PM_{10}$ ratio
Test Site Criteria (B)	Number of test sites	2	2	No change in recommendations
	Testing site locations	Two (2) different climate regions across the U.S. that <u>are not adjacent</u>	Two (2) different climate regions across the U.S. that <u>can be adjacent</u>	Testing locations that meet the test site criteria may not be as widespread for evaluating PM_{10} sensors. Two locations impacted by different sources (e.g., windblown dust, industrial coarse PM) would be most informative
	Goal 24-hour average concentration at test site	≥ 25 $\mu g/m^3$ for at least one day	≥ 40 $\mu g/m^3$ for at least one day	A higher goal concentration is recommended because PM_{10} concentrations are typically higher
Data averaging timeframe (B)		24-hours	24-hours	No change in recommendations
$PM_{2.5}/PM_{10}$ Ratio (B, E)		N/A	Goal $PM_{2.5}/PM_{10}$ Ratio of ≤ 0.4 for at least 1 hour	Ratio suggested to understand how well the sensor measures PM_{10}
Test aerosols (E)		Ammonium sulfate, ammonium nitrate, oleic acid or sucrose, smoke from smoldering wood chips or other biomass material	Arizona test dust, dolomite powder, fine test dust, coarse test dust	Test aerosols that are larger in size are needed to simulate PM_{10} aerosols

All testers are encouraged to conduct base testing at a minimum. Enhanced testing is also encouraged although it calls for a controlled laboratory exposure chamber. For NSIM applications where high PM_{10} concentrations are expected (e.g., dust storms, construction), it is recommended that testers conduct base testing in more than two locations and include sites impacted by dust sources, industrial sources, and/or higher PM_{10} concentrations.

Performance metrics and corresponding target values have been identified based on the current state-of-the-science, literature reviews, findings from other organizations that conduct routine sensor evaluations, sensor standards/certification programs in development by other organizations, and the Air Quality Sensor Performance Evaluation Center (AQ-SPEC) and U.S. EPA expertise in sensor evaluation research. A summary of the performance metrics and target values for the base and enhanced testing protocols are shown in Table ES-2. Information for $PM_{2.5}$ sensors is included for comparison only. For base testing of PM_{10} sensors, an additional data visualization called 'exploring meteorological effects' is recommended which includes graphing meteorological data to understand its influences on sensor performance. Further for base testing, it is recommended that at the base test sites, at least one day has a 24-hour average PM_{10} concentration of at least 40 micrograms per cubic meter (μg/m^3) or greater, and a $PM_{2.5}/PM_{10}$ ratio of less than or equal to 0.4 for at least one hour during the testing period. Additional details can be found in Section 2.0 in this report.

Table ES-2.Base and Enhanced Testing – Recommend Performance Metrics and Target Values for $PM_{2.5}$ and PM_{10} Air Sensors

Performance Metric		Target Value		
		$PM_{2.5}$ Base Testing	**PM_{10} Base Testing**	**Enhanced Testing**
Precision	Standard Deviation (SD) -OR- Coefficient of Variation (CV)	≤ 5 μg/m^3	≤ 5 μg/m^3	No target values recommended; report results
		≤ 30%	≤ 30%	
Bias	Slope	1 ± 0.35	1 ± 0.35	
	Intercept (b)	-5 ≤ b ≤ 5 μg/m^3	-10 ≤ b ≤ 10 μg/m^3	
Linearity	Coefficient of Determination (R^2)	≥ 0.70	≥ 0.70	
Error	Root Mean Square Error (RMSE) or Normalized Root Mean Square Error (NRMSE)	RMSE ≤ 7 μg/m^3 or NRMSE ≤ 30%†	RMSE ≤ 14 μg/m^3 or NRMSE ≤ 30%†	

†A sensor will meet this target if either the RMSE or NRMSE meet this criterion. See Appendix D for further discussion.

Except for the $PM_{2.5}/PM_{10}$ ratio performance metric, instructions on how to calculate the other performance metrics can be found in Section 3.0 (*Performance Metrics and Supporting Calculations for Evaluating $PM_{2.5}$ Air Sensors*) of the $PM_{2.5}$ Targets Report.

The performance metrics and target values for base and enhanced testing are recommended based on the current knowledge of PM_{10} air sensors at the time this report was released. Target values for enhanced testing are not included at this time due to limited feasibility, lack of consensus regarding testing protocols, and inconsistency in sensor evaluation results that can result due to the limited amount of data that will be collected and variation in the tester's choice of PM surrogate.

Test conditions for the enhanced testing protocols are shown in Table ES-3. Information for $PM_{2.5}$ sensors is included for comparison only. Further details on the PM_{10} enhanced testing protocols can be found in Section 2.2.

Table ES-3. Enhanced Testing – Additional Recommended Performance Metrics and Test Conditions for $PM_{2.5}$ and PM_{10} Air Sensors

Performance Metric	Test Conditions		
	Parameter	**$PM_{2.5}$**	**PM_{10}**
Effect of Relative Humidity (RH)	Moderate RH	40% ± 5%	40% ± 5%
	Elevated RH	85% ± 5%	85% ± 5%
Effect of Temperature (T)	Moderate T	20°C ± 1°C	20°C ± 1°C
	Elevated T	40°C ± 1°C	40°C ± 1°C
Drift	Low concentration	10 µg/m^3 ± 10%	15 µg/m^3 ± 10%
	Mid concentration	35 µg/m^3 ± 5%	60 µg/m^3 ± 5%
Accuracy at High Concentrations	High concentration	150 µg/m^3 ± 5%	200 µg/m^3 ± 5%
	Higher concentration	250 µg/m^3 ± 5%	300 µg/m^3 ± 5%

Because $PM_{2.5}$ and PM_{10} sensors are often used in similar outdoor environments, the temperature (T) and relative humidity (RH) test conditions remain the same. The mid, high, and higher concentration levels for $PM_{2.5}$ enhanced testing were set at the Air Quality Index (AQI) level breakpoints. The concentration levels for PM_{10} enhanced testing were similarly set at corresponding breakpoints for the PM_{10} AQI levels. Templates for base and enhanced testing reports to aid in the consistent reporting of testing results can be found in Appendices F and H. Testing results do not constitute certification or endorsement by the U.S. EPA. It is recommended that testers make the testing reports available on their respective websites to inform consumers on the testing results.

1.0 Introduction

1.1 Background

Air sensors continue to be more accessible nationwide, and their development is evolving at a rapid pace. The use of air sensors is more widespread for a variety of air quality monitoring applications and data sets are readily available to the public. While air sensors continue to encourage innovation in air quality monitoring, the data quality is still highly variable. In response to the widespread use of air sensor technologies, variability in sensor data quality, and the lack of consistent testing protocols to evaluate sensor performance, the United States Environmental Protection Agency (U.S. EPA) published a report in 2021 outlining recommended testing protocols to evaluate the performance of fine particulate matter ($PM_{2.5}$) air sensors for non-regulatory supplemental and informational monitoring applications (NSIM) in ambient, outdoor, fixed site environments (U.S. EPA, 2021; hereafter referred to as the "$PM_{2.5}$ Targets Report").

NSIM categories and examples are as follows (*from $PM_{2.5}$ Targets Report, Table 1-1*):

- **Spatiotemporal Variability**: Characterizing a pollutant concentration over a geographic area and/or time (e.g., daily trends, gradient studies, air quality forecasting, participatory/citizen science, education)

- **Comparison**: Analysis of differences and/or similarities in air pollution characteristics against a threshold value of between different networks, locations, regions, time periods, etc. (e.g., hotspot detection, data fusion, emergency response, supplemental monitoring)

- **Long-term Trend:** Change in a pollutant concentration over a period of typically years (e.g., long-term changes, epidemiological studies, model verification)

In addition to the testing protocols, the $PM_{2.5}$ Targets Report includes recommended performance metrics and corresponding target values (testable performance metrics) to understand and compare sensor performance. Reporting templates are also included to encourage consistent communication and display of performance evaluation results.

The U.S. EPA initially provided guidance on evaluating $PM_{2.5}$ sensors as there was more widespread use and more technical information and knowledge available for these technologies. Often many sensors that measure $PM_{2.5}$ also report concentrations for inhalable particles with diameters that are generally less 10 micrometers (PM_{10}). However, information on the ability of sensors to properly measure PM_{10} is more uncertain. There are several known measurement challenges with existing PM_{10} sensors (Mukherjee et al., 2017; Jayaratne et al., 2018; Rogulski and Badyda, 2020; Tagle et al., 2020; Qin et al., 2020; Duvall et al., 2021; Vogt et al., 2021) as summarized below:

- **Selectivity to Particle Size**: Sensors may not measure the target pollutant accurately and reliably within the expected concentration range for the application; particle sensors often measure a range of particle diameters which may only represent a portion of the PM_{10} diameter range;

- **Meteorological and Composition Effects**: Relative humidity (RH), temperature (T), and variations in particle composition or size can impact measurements;

- **Poor Aspiration Efficiency**: Low flowrates, common in currently available PM_{10} sensors due to use of small fans or pumps, may not properly sample larger particles which normally require a large flowrate; aspiration efficiency can also be influenced by ambient wind speed and wind direction;

- **Design and Air Flow**: Features such as the positioning of a device's sampling train, shape and size of the sampling inlet, and bends within the sampling airflow path can impact the collection efficiency and measurement of PM_{10};

- **Drift**: A sensor's response can change over time and readings can become inaccurate or unreliable, such as due to dust deposition or light source degradation; and

- **Data Correction**: Sensors may perform poorly out-of-the box and correction algorithms may be needed to provide more accurate data.

Despite these challenges, there is interest in the use of PM_{10} sensors for a variety of NSIM applications. As identified in the U.S. EPA's 2019 "Deliberating Air Quality Sensor Targets Workshop" (Duvall et al., 2021), desired uses of PM_{10} sensors include characterizing ambient air, hotspots, dust sources (e.g., storms, construction, agriculture), and fires.

1.2 Motivation

The motivation for this report is similar to the $PM_{2.5}$ Targets Report (*see Section 1.2 of that report*). Consistent testing protocols are needed that can be used for uniform evaluation and comparison of different PM_{10} sensor technologies. Additionally, recommended and testable performance metrics (i.e., performance targets) for PM_{10} sensors that can distinguish performance among different devices do not currently exist.

The use cases of interest for PM_{10} sensors, are largely focused on understanding PM_{10} concentrations near a source or in locations which experience higher than typical ambient PM_{10} levels. To ensure that testing is conducted in an appropriate testing environment, this report provides guidance and education on testing locations.

While PM_{10} air sensors are commercially available and being used, fewer performance evaluations and successful field applications have been reported suggesting a need for further development, testing, and improvements. Despite these limitations, the U.S. EPA recognizes that consumers are interested in using PM_{10} sensors for a variety of NSIM applications. Testing protocols, performance metrics, and target values can be useful for evaluating the performance of PM_{10} sensors and helping guide future technology development and improvements.

1.3 Objectives

The purpose of this report is to provide a standard, streamlined, unbiased approach to testing the performance of PM_{10} air sensors for NSIM applications in ambient, outdoor,

fixed site environments. Given different potential use cases and testing considerations for PM_{10} sensors, this report provides more guidance on how to appropriately conduct testing. Specific objectives of this report are as follows:

- Provide a consistent set of testing protocols, metrics, and target values to systematically evaluate the performance of air sensors.
- Provide guidance on suitable applications, measurement considerations, appropriate measuring locations, and expected PM_{10} concentration ranges in the ambient, outdoor environment.
- Provide a consistent framework for communicating performance evaluation results.
- Help consumers make informed decisions on choosing sensors that might best suit a NSIM application of interest.

It should be noted that other applications (e.g., mobile monitoring, indoor monitoring, personal exposure monitoring) may require different testing protocols which are not covered in this report.

The PM_{10} supplemental report should be used as a complement to the $PM_{2.5}$ Targets Report. A supplemental approach was selected as there is significant overlap with the $PM_{2.5}$ Targets Report. Use of this format will help highlight new information and the key differences and considerations for evaluating PM_{10} air sensors.

Testers should carefully read and follow the stepwise approach for the following sections in the $PM_{2.5}$ Targets Report:

- **Section 2.0** "Performance Testing Protocols for $PM_{2.5}$ Air Sensors"
- **Section 3.0** "Performance Metrics and Supporting Calculations for Evaluation $PM_{2.5}$ Air Sensors"

Where indicated in this supplement, changes or additions to the testing protocols including special considerations, additional performance metric calculations or visualizations, and updated target values for evaluating PM_{10} sensors are provided.

The intended audience for this report includes potential testing organizations, sensor manufacturers, and sensor developers. It is anticipated that a variety of consumers (e.g., state/local/tribal agencies, federal government agencies, community groups, participatory/citizen scientists, academia) will benefit from the consistent presentation of testing results to identify sensor technologies that would be best suited for their NSIM application and understand the performance of the air sensor technologies. Consumers may also choose to conduct these testing protocols.

As with the $PM_{2.5}$ Targets Report, results from the PM_{10} sensor testing protocols do not constitute certification or endorsement by the U.S. EPA. It is recommended that testers make the testing reports available on their respective websites to inform consumers on the testing results.

2.0 Performance Testing Protocols for PM_{10} Air Sensors

The procedures outlined in this section provide standardized test protocols for evaluating the performance of PM_{10} air sensors. Testers should refer to the $PM_{2.5}$ Targets Report introductory paragraphs in Section 2.0 (*Performance Testing Protocols for $PM_{2.5}$ Air Sensors*) for a general summary of the recommended testing procedures. Figure 2-1 provides an example approach for conducting base and enhanced testing with a single set of air sensors.

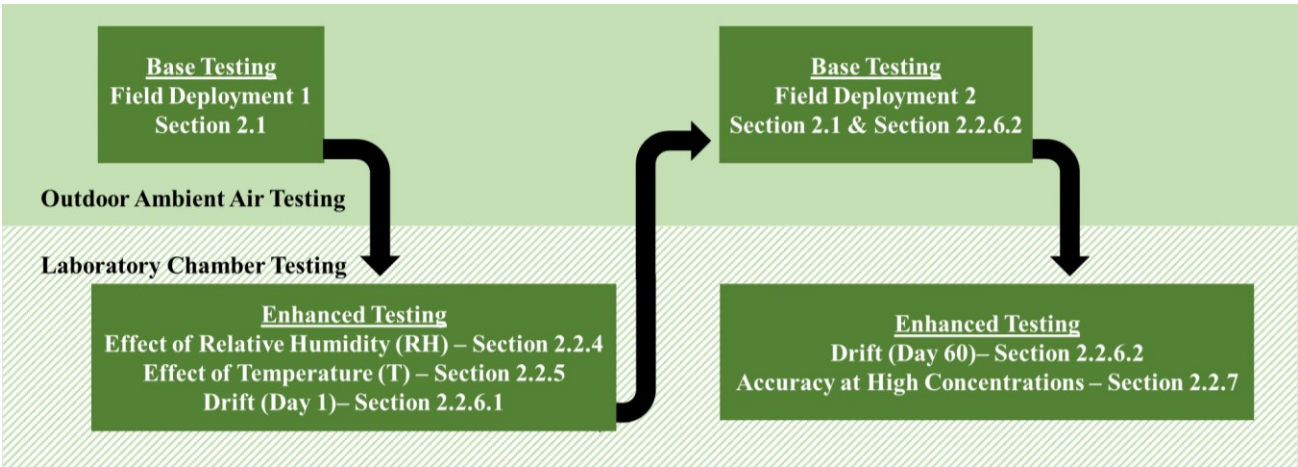


Figure 2-1. Example Approach for Conducting Base and Enhanced Testing of a Single Set of Air Sensors

2.1 Base Testing

Briefly, base testing consists of two (2) field deployments of PM_{10} air sensors with collocated Federal Reference or Federal Equivalent Method (FRM/FEM) monitors for at least 30 days for each deployment. Two deployment locations impacted by different sources (e.g., windblown dust, industrial coarse PM) would be most informative for testing. As discussed further in Section 2.1.1, testers may also wish to operate additional reference monitors to understand particle size distribution and composition which may impact sensor performance.

Testers may set up their own FRM/FEM monitors using guidance and information on ambient air monitoring and monitoring methods available on the U.S. EPA's Ambient Air Monitoring and Technology Information Center (AMTIC) air monitoring methods webpage (https://www.epa.gov/amtic/air-monitoring-methods, *last accessed 09/14/2023*). FRM/FEM and reference monitor flow rates should be checked, and temperature and pressure sensors should be calibrated and checked using transfer standards that are certified and traceable by the National Institute of Standards and Technology (NIST) standards. Alternatively, testers may wish to develop relationships with state, local, or Tribal environmental agencies to collocate sensors near FRM/FEM monitors at existing air quality monitoring sites around the U.S. These sites can be found on the following website:

https://www.epa.gov/outdoor-air-quality-data/interactive-map-air-quality-monitors (*last accessed 05/30/2023*).

Base testing should occur at two (2) different sites in two (2) different U.S. climate regions impacted by different sources to give a variety of particle sources, types, and size distributions. The combination of field tests should demonstrate sensor performance over a range of T, RH, and weather conditions that will inform on sensor use across the U.S. For base testing, the sensor and FRM/FEM or reference data should be compared at 24-hour averages (with 75% data completeness) As there is little variation between different types of FRM/FEM monitors at this scale per federal requirements. It is acknowledged that a paired sensor and FRM/FEM reference instrument may be capable of providing higher time resolution data (e.g., 1-hour averaged measurements). Testers may produce additional testing reports reflecting performance at additional time averaging intervals. It is also acknowledged that a sensor device may have the capability to measure both $PM_{2.5}$ and PM_{10}. Prospective testers may conduct base testing for both $PM_{2.5}$ and PM_{10} sensors at the same time if the test site selection criteria are met for each PM size fraction (see Section 2.1.2 in both this report and the $PM_{2.5}$ Targets Report).

There are several important testing considerations that are unique to PM_{10} which are briefly summarized below.

Particle Size Distribution: PM_{10} includes both the coarse particulate matter (PMc) and $PM_{2.5}$ size fractions. Coarse PM (also known as coarse particles, PMc, or $PM_{10-2.5}$) are particles with diameters generally larger than 2.5 micrometers (µm) and smaller than, or equal to,10 µm in diameter. $PM_{2.5}$ have particle sizes generally 2.5 µm in diameter or smaller. Examining these size fractions can provide important information on the dominant particle sizes in ambient air and potential sources of particulate matter (PM).

$PM_{2.5}/PM_{10}$ ratios can be used as an indicator of the size distribution of PM in ambient air – lower ratios indicate that larger particles are more dominate. Testing in conditions where $PM_{2.5}/PM_{10}$ ratios vary can inform whether PM_{10} sensors are measuring coarse PM. The PM_{10} testing protocols recommend a goal 1-hour $PM_{2.5}/PM_{10}$ ratio during the testing period to ensure that the sensors are experiencing ambient, outdoor conditions that are impacted by larger particles and/or sources that generate larger particles. Testing under these specific conditions will better inform consumers on how PM_{10} sensors perform in ambient, outdoor air.

PM_{10} Testing Locations: The base testing protocols recommend that testers select two (2) test sites with a goal 24-hour average PM_{10} concentration. Ambient levels of PM_{10} are likely to be higher in certain regions of the U.S. (e.g., Southwest, Northwest, West), during dust storms, and near dust sources such as construction, agriculture operations, and quarries. Testers should consider sites and/or regions that are likely to experience higher PM_{10} concentrations. Additional information on PM_{10} concentrations is provided in Section 2.1.2 and Appendix B in this report.

Use of Existing Ambient Monitoring Sites: Monitors in ambient air monitoring networks have specific purposes and those purposes dictate how monitors are sited. For example, monitors used to determine regulatory compliance are sited based on population density, peak concentration of the pollutant in a given area, pollution source, background levels of

the pollutant, and transport of the pollutant. Existing ambient monitoring network sites may not be impacted by significant PM_{10} sources or experience the goal 24-hour average PM_{10} concentration or the goal 1-hour $PM_{2.5}/PM_{10}$ ratio recommended in these test protocols. Therefore, testers may need to set up their own test sites. Please refer to Section 2.1.2 (*Selecting and Setting up a Test Site*) in the $PM_{2.5}$ Targets Report for guidance on setting up testing locations.

PM_{10} FEM Designations: Some $PM_{2.5}$ FEM monitors also measure PM_{10}, however the monitor may not be designated as an FEM for PM_{10}. Given the importance of the $PM_{2.5}/PM_{10}$ ratio for evaluating PM_{10} sensors, care should be taken to ensure that the monitor(s) used for base testing is FEM-designated for $PM_{2.5}$ and PM_{10}. Separate $PM_{2.5}$ and PM_{10} monitors could be operated concurrently.

Table 2-1 highlights the key similarities and differences for base testing of $PM_{2.5}$ and PM_{10} air sensors. The procedure in this section outlines the materials and equipment needed, site selection, setup, testing procedure, and data reporting needs to evaluate PM_{10} air sensor performance. Please refer to the $PM_{2.5}$ Targets report as noted. To assist testers in ensuring that the requested data before and during the base testing procedure is documented, a checklist is provided in Appendix E. All information for this testing procedure should be recorded in the base testing report (template available in Appendix F). As previously mentioned, it is recommended that testers make the testing report(s) available on their respective websites to inform consumers on the testing results.

Table 2-1. Summary of the Key Similarities and Differences for Base Testing of $PM_{2.5}$ and PM_{10} Air Sensors (gray boxes indicate differences)

Category	$PM_{2.5}$	PM_{10}	Comments
Number of Sensors	Three (3) sensors of the same make, model and firmware version	Three (3) sensors of the same make, model and firmware version	No change in recommendations
FRM/FEM Monitors	Calibrated $PM_{2.5}$ FRM/FEM monitor	Calibrated $PM_{2.5}$ FRM/FEM monitor AND a calibrated PM_{10} FRM/FEM monitor	Both a calibrated $PM_{2.5}$ and PM_{10} FRM/FEM monitor are needed to calculate a $PM_{2.5}/PM_{10}$ ratio
Number of test sites	2	2	No change in recommendations
Testing site locations	Two (2) different climate regions across the U.S. that are not adjacent	Two (2) different climate regions across the U.S. that can be adjacent	Testing locations that meet the test site criteria may not be as widespread for evaluating PM_{10} sensors. Two locations impacted by different sources (e.g., windblown dust, industrial coarse PM) would be most informative.
Averaging timeframe	24-hours	24-hours	No change in recommendations
Goal 24-hour average concentration at test site	≥ 25 µg/m^3 for at least one day	≥ 40 µg/m^3 for at least one day	A higher goal concentration is recommended because PM_{10} concentrations are generally higher
$PM_{2.5}/PM_{10}$ Ratio	N/A	Goal $PM_{2.5}/PM_{10}$ Ratio of ≤ 0.4 for at least one hour	Ratio suggested to understand how well the sensor measures PM_{10}

2.1.1 Materials and Equipment

The following materials and equipment are needed for this testing procedure:

- Three (3) or more PM_{10} air sensors having the same make, model, and firmware version
- Calibrated PM_{10} FRM/FEM monitor[*, #]
- Calibrated $PM_{2.5}$ FRM/FEM monitor[*, #, **]
- Calibrated RH monitor[†]

- Calibrated T monitor†

- Support structures and/or enclosures for air sensors (as recommended by the manufacturer)

*The FRM/FEM monitor must be calibrated on site prior to conducting base testing. Additional materials may be needed to accomplish the calibration. Calibration procedures are outlined in 40 CFR Parts 50, 53, and Appendix A of Part 58. Calibration procedures for continuous FEMs are detailed in the manufacturer's user manual which are approved as part of the FEM designation process. If testing is conducted at an established air quality regulatory monitoring station with established calibration and quality control procedures, attach or cite the Quality Assurance Project Plan (QAPP) to the base testing report (see Appendix F). Additionally, if testing at an established site, it is recommended to confirm with the site operators whether the FRM/FEM monitor(s) passed the monthly checks before and after testing the air sensors and include this information in the base testing report.

#While some designated $PM_{2.5}$ FEMs also measure PM_{10}, the monitors may not be designated as an FEM for measurement of PM_{10}. Testers should visit the U.S. EPA's AMTIC air monitoring methods webpage for a list of designated PM_{10} FEM monitors (https://www.epa.gov/amtic/air-monitoring-methods, *last accessed 10/24/2023*).

**Calibration procedures are also outlined in the Quality Assurance Guidance Document 2.12 (U.S. EPA, 2016; herein referred to as QA Document 2.12).

†Meteorological monitors should be certified by the manufacturer or calibrated, install, maintained, and audited according to quality assurance procedures outlined in U.S. EPA's Quality Assurance Handbook for Air Pollution Measurement Systems Volume IV: Meteorological Measurements (U.S. EPA, 2008)

Testers should carefully review Section 2.1.1 (*Materials and Equipment*) in the $PM_{2.5}$ Targets Report for information on additional useful measurements (e.g., particle size distribution, particle chemical composition, refractive index), logging data, time aligning data, and data averaging.

2.1.2 Selecting and Setting up a Test Site

Testing in locations with higher PM_{10} concentrations and different temperature and humidity conditions is most informative for understanding PM_{10} sensor performance. Table 2-2 provides recommended criteria for the PM_{10} test sites. For U.S. climate regions, please refer to the $PM_{2.5}$ Targets Report (*see Figure 2-2).*

Table 2-2. Test Site Selection Criteria for PM_{10} Sensors

Base Testing Plan	Location(s)	Season	Goal 24-Hour Average PM_{10} Concentration (for at least one day)	$PM_{2.5}/PM_{10}$ Ratio (for at least one hour)
Two Test Sites	Site 1	Climate Region 1	≥ 40 µg/m^3	≤ 0.4
	Site 2	Climate Region 2	≥ 40 µg/m^3	≤ 0.4

A goal of at least one day of the 30-day testing period with a 24-hour average PM_{10} concentration of at least 40 µg/m^3 is suggested (refer to Appendix B for more detailed information on this recommended testing criteria). Ambient levels of PM_{10} are likely to be higher in certain regions of the U.S. (e.g., Southwest, Northwest, West; see Table 2-3 for annual trends). Higher ambient PM_{10} concentrations also occur during dust storms and near dust sources such as construction, agriculture operations, and quarries. It is acknowledged that PM_{10} concentrations across the U.S. may be unpredictable and even sites that are expected to exceed 40 µg/m^3 may not achieve this level. A level of 40 µg/m^3 is recommended as a goal as this will help ensure statistics are comparable across sites and that testing does not result in a low R^2 due to low PM_{10} concentration ranges. If this goal is not achieved, note this as a limitation in the testing report.

Table 2-3. U.S. Regional PM_{10} Trends from 2018-2022 (Source: National Air Quality Status and Trends of Key Air Pollutants; available at *https://www.epa.gov/air-trends*, last accessed 05/30/2023)

U.S. Region	PM_{10} Concentration (µg/m^3)*		
	Minimum	Maximum	Average
Central (Ohio Valley)	44.79	55.75	49.94
Northeast	35.87	42.53	38.81
Northern Rockies and Plains	49.32	86.06	71.41
Northwest	60.80	206.00	119.76
South	44.54	75.38	66.86
Southeast	36.05	45.06	42.82
Southwest	64.86	127.97	101.12
Upper Midwest	48.00	57.55	52.95
West	76.98	115.44	97.91

*Values represent the minimum, maximum, and average of all sites within the U.S. climate region using the annual 2nd maximum 24-hour averaged PM_{10} concentration. These data include concentrations that may have been impacted by episodic events (e.g., wildfires, dust storms).

Based on historical data, there are a number of sites across the U.S. that could offer the conditions shown in Table 2-3. If using an existing ambient air monitoring network site [e.g., National Core Network (NCore), State and Local Air Monitoring Station (SLAMS)], a tester can examine historical air quality data, found on the U.S. EPA AirData website (https://www.epa.gov/outdoor-air-quality-data, *last accessed 04/19/2023*), to see if/when a site is likely to meet the criteria. If using another site that does not have historical air quality data, consult data from the nearest regulatory monitoring site to determine if the site(s) is likely to meet the criteria in Table 2-2.

A $PM_{2.5}/PM_{10}$ ratio is recommended to understand the size distribution of PM at the test site(s). Conditions where $PM_{2.5}/PM_{10}$ ratios vary can better inform whether PM_{10} sensors are measuring the coarse PM size fraction. A goal $PM_{2.5}/PM_{10}$ ratio of ≤ 0.4 for at least one hour during the 30-day testing period is suggested (more information on these thresholds can be found in Appendix B).

Follow Steps 1 to 3 in Section 2.1.2 (*Selecting and Setting up a Test Site*) in the $PM_{2.5}$ Targets Report. Testers should use the test site criteria from this report (Table 2-2) and the PM_{10} base testing reporting template provided in this report (see Appendix F).

2.1.3 Setting up the Air Sensors

Follow Steps 1 to 5 in Section 2.1.3 (*Setting up the Air Sensors*) in the $PM_{2.5}$ Targets Report. PM_{10} sensors should be used. It is recognized that a sensor device may have the capability to measure both PM_{10} and $PM_{2.5}$. Prospective testers may conduct base testing for both $PM_{2.5}$ and PM_{10} sensors at the same time if the test site selection criteria are met for each PM size fraction. Testers should use the PM_{10} base testing reporting template provided in this report (see Appendix F).

2.1.4 Conduct Base Testing

Follow Steps 1 to 11 in Section 2.1.4 (*Conduct Base Testing*) in the $PM_{2.5}$ Targets Report. In addition to the $PM_{2.5}$ FRM/FEM monitor, a PM_{10} FRM/FEM, and PM_{10} sensors should be used. It is recognized that a sensor device may have the capability to measure both PM_{10} and $PM_{2.5}$. Note that the QA Document 2.12 only applies to $PM_{2.5}$ FEM monitors. Testers should use the test site criteria from this report (Table 2-2) and the PM_{10} base testing reporting template provided in this report (see Appendix F).

Base testing should be conducted for 30 consecutive days. It is acknowledged that instrument failure or site issues could cause loss of data and an interrupted test. Testing can be continued or restarted to achieve the 30-day deployment goal. For further discussion refer to the Section 2.1.4 in the $PM_{2.5}$ Targets Report.

2.2 Enhanced Testing

Testers should refer to Section 2.2 (*Enhanced Testing*) in the $PM_{2.5}$ Targets Report for a general overview and important information related to the enhanced testing procedures. Figure 2-2 shows an overview of the enhanced testing procedures.

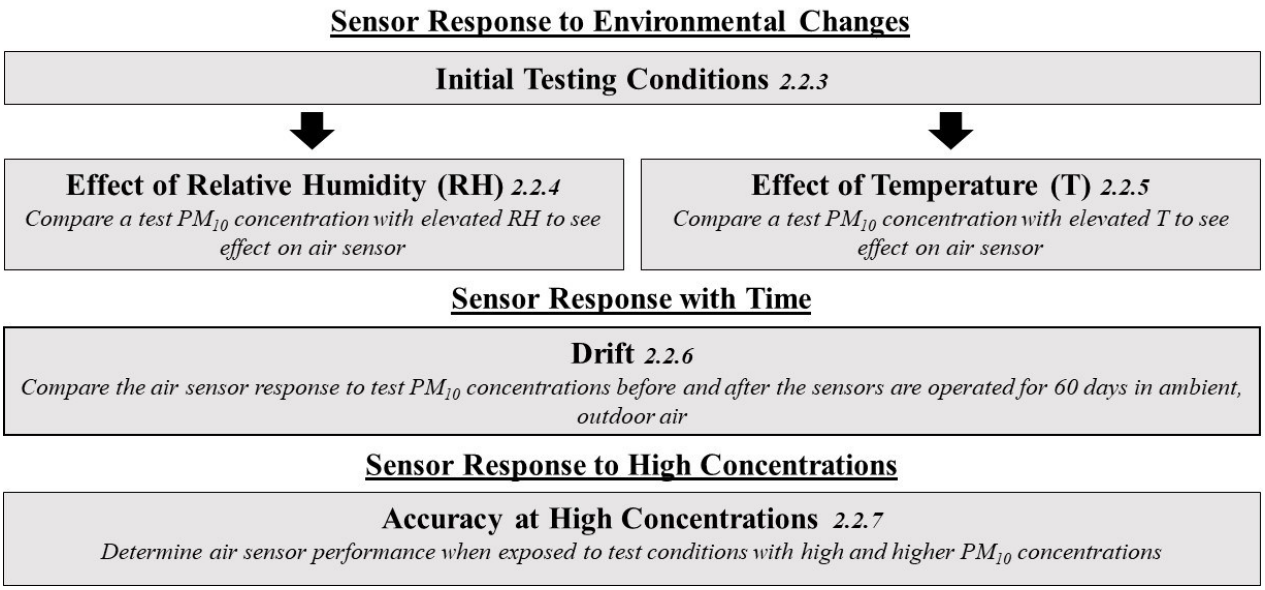


Figure 2-2. Overview of the Enhanced Testing Procedures for PM_{10} Sensors

Briefly, enhanced testing consists of testing air sensors in a controlled laboratory environment to understand the effects of RH and T, and other important parameters including drift and measurement accuracy at higher concentrations levels. Enhanced testing also allows sensors to be tested at concentrations that may be rarely encountered in the field yet important to understand. Although the T and RH test conditions are the extent of these protocols, we acknowledge that additional testing at higher T and RH levels may be valuable considering the environments where PM_{10} sensors are deployed. Testers may choose to conduct additional enhanced testing under different T and RH conditions.

Given variations in PM_{10} concentration, particle composition, and properties of particles (e.g., optical characteristics, potential health effects), it is impossible to generate an aerosol laboratory that would perfectly simulate those encountered under field conditions. Therefore, testers may prefer to conduct additional base testing (field deployments) rather than enhanced testing (laboratory testing).

There are several important considerations for enhanced testing of PM_{10} sensors which are briefly summarized below.

Test Aerosol: Test aerosols that are larger in size are needed to simulate PM_{10} conditions and particle sizes. At this time, there is no consensus among organizations about which test aerosols would be most informative for use in the enhanced testing, so no recommendations have been made at this time. Test aerosols that may be reasonable proxies for PM_{10} include:

- Arizona test dust
- Dolomite
- Fine test dust (ISO 12103-1, A2)*
- Coarse test dust (ISO 12103-1, A4)*

*ISO = International Organization for Standardization

To the extent possible, NIST or ISO certified test aerosol should be used. The type of particles generated for the tests and their particle size distribution, chemical composition, and refractive index, if known, should be recorded on the enhanced testing report. It should be noted that the type of particles dispersed can influence the results of the enhanced test.

Particle Size Distribution: Testers should carefully consider particle size. As mentioned previously, PM_{10} includes both the $PM_{10-2.5}$ (coarse) and $PM_{2.5}$ (fine) size fractions. It is important to test whether the sensor can measure both the coarse and fine aerosol size fractions by testing each separately or by using a test dust that contains a wide range of particle sizes. If using the latter approach, a range of $PM_{2.5}/PM_{10}$ ratios is preferred with a goal of meeting a ratio of ≤ 0.4 for a portion of the test is recommended. Additional equipment is suggested to check that the test aerosols are the appropriate size to understand the performance of PM_{10} sensors.

PM_{10} FEM Designations: Some $PM_{2.5}$ FEM monitors also measure PM_{10}, however the monitor may not be designated as an FEM for PM_{10}. Given the importance of the $PM_{2.5}/PM_{10}$ ratio for evaluating PM_{10} sensors, care should be taken to ensure that the monitor(s) used for enhanced testing is FEM-designated for $PM_{2.5}$ and PM_{10}. Separate $PM_{2.5}$ and PM_{10} monitors could be operated concurrently. Table 2-4 highlights the key similarities and differences for enhanced testing of $PM_{2.5}$ and PM_{10} air sensors. To assist testers in ensuring that the requested data before and during the enhanced testing procedure is documented, a checklist is provided in Appendix G of this report. All information for this testing procedure should be recorded in the enhanced testing reporting template (Appendix H in this report). As mentioned previously, it is recommended that testers make the testing report(s) available on their respective websites to inform consumers.

Table 2-4. Summary of the Key Similarities and Differences for Enhanced Testing of $PM_{2.5}$ and PM_{10} Air Sensors (*gray shaded boxes indicate differences*)

Category	$PM_{2.5}$	PM_{10}	Comments
Sensors	Three (3) sensors of the same make, model, and firmware version	Three (3) sensors of the same make, model, and firmware version	No change in recommendations
FEM monitors	Calibrated $PM_{2.5}$ FEM monitor	Calibrated $PM_{2.5}$ FEM monitor AND a calibrated PM_{10} FEM monitor	Both a calibrated $PM_{2.5}$ and PM_{10} FEM monitor are recommended to estimate a $PM_{2.5}/PM_{10}$ ratio
Particle dispenser	N/A	Particle dispenser system	Dispenses larger particles for measurement in the exposure chamber
$PM_{2.5}/PM_{10}$ Ratio	N/A	Goal $PM_{2.5}/PM_{10}$ Ratio of ≤ 0.4	Ratio suggested to understand how well the sensor measures PM_{10}
Test Aerosol	Ammonium sulfate, ammonium nitrate, oleic acid or sucrose, smoke from smoldering wood chips or other biomass material	Arizona test dust, dolomite powder, fine test dust, coarse test dust	Larger particles are needed to evaluate PM_{10} sensors

2.2.1 Materials and Equipment

The following materials and equipment are needed for this testing procedure:

- Three (3) or more PM_{10} air sensors having the same make, model, and firmware version[*]
- Calibrated PM_{10} FEM monitor[#, **]
- Calibrated $PM_{2.5}$ FEM monitor[#, **]
- Exposure chamber that can control environmental conditions
- Particle dispenser system
- Aerodynamic particle sizing instrument
- Zero air generator

- Calibrated RH monitor†

- Calibrated T monitor†

*Sensors can be the same ones used in the base testing procedure.

#The FEM monitor must be calibrated on-site prior to conducting enhanced testing and additional materials and equipment may be needed to accomplish the calibration. Calibration procedures are outlined in the manufacturer's user manual which are approved as part of the FEM designation process. For $PM_{2.5}$ continuous monitors, calibration procedures are also outlined in QA Document 2.12. If testing is conducted at an established sensor testing facility with established calibration and QC procedures, attach or cite the QAPP in the enhanced testing report (Appendix H).

**While some designated $PM_{2.5}$ FEMs also measure PM_{10}, the monitors may not be designated as an FEM for measurement of PM_{10}. Testers should visit the U.S. EPA's AMTIC air monitoring methods webpage for a list of designated PM_{10} FEM monitors (https://www.epa.gov/amtic/air-monitoring-methods, *last accessed 10/24/2023*).).

†Meteorological monitors should be certified by the manufacturer or calibrated, installed, maintained, and audited according to quality assurance procedures outlined in U.S. EPA's Quality Assurance Handbook for Air Pollution Measurement Systems Volume IV: Meteorological Measurements (U.S. EPA, 2008)

Testers should carefully review Section 2.2.1 (*Materials and Equipment*) in the $PM_{2.5}$ Targets Report for information on additional useful measurements (e.g., particle size distribution, particle chemical composition, refractive index), logging data, time aligning data, data averaging, and criteria for the exposure chamber.

In addition to the criteria listed in the $PM_{2.5}$ Targets Report, the exposure chamber should also meet these criteria:

- Generate and measure the recommended PM_{10} concentrations within the range of 5-300 µg/m^3 based on the testing conditions outlined in this supplement report.

- The particle dispenser system should be positioned above the exposure chamber and connected with as few bends as possible to prevent particle loss and build-up before particles enter the chamber.

- Particle sizer should be capable of measuring aerodynamic particle size ranging from 0.5 to 20 µm.

- The PM_{10} FEM monitor and particle sizer instrument should be mounted directly below the sampling chamber to reduce particle loss in the sampling line due to bends.

If possible, provide documentation on the chamber specifications, characterization, and any laboratory intercomparison(s).

2.2.2 Equipment Set Up in Exposure Chamber

Follow Steps 1 to 9 in Section 2.2.2 (*Equipment Set Up in Exposure Chamber)* in the $PM_{2.5}$ Targets Report. A PM_{10} FEM, $PM_{2.5}$ FEM, and PM_{10} sensors should be used. It is recognized that a sensor device may have the capability to measure both $PM_{2.5}$ and PM_{10}. Note that QA Document 2.12 only applies to $PM_{2.5}$ FEM monitors.

2.2.3 Initial Testing Conditions

Take the steps below to begin the enhanced testing procedure.

1. Supply the exposure chamber with all conditions shown in Table 2-5.
2. Allow all measurements to stabilize within all tolerances shown in Table 2-5.
3. Once steady state is achieved, collect either a minimum of 20-30 pairs of time-matched sensor and FEM data points or three (3) consecutive hours for the parameters listed below.
    a. PM_{10} concentration from each sensor (µg/m^3)
    b. FEM PM_{10} concentration (µg/m^3)
    c. FEM $PM_{2.5}$ concentration (µg/m^3)
    d. $PM_{2.5}/PM_{10}$ ratio
    e. T (°C)
    f. RH (%)

Table 2-5. Initial Testing Conditions for PM_{10} Sensors

Parameter	Reference Setpoint
PM_{10} Concentration	60 µg/m^3 ± 5%
$PM_{2.5}/PM_{10}$ Ratio	≤ 0.4 ± 0.1
T	20°C ± 1°C
RH	40% ± 5%

2.2.4 Effect of Relative Humidity (RH)

To determine the effect of elevated RH on sensor performance, take the steps below.

1. Repeat the procedure outlined in Section 2.2.3 of this report.
2. Supply the exposure chamber with the conditions in Table 2-6.
3. Allow the measurements to stabilize within all tolerances shown in Table 2-6.
4. Once steady state is achieved, collect either a minimum of 20-30 pairs of time-matched sensor and FEM data points or three (3) consecutive hours for the parameters listed below.
    a. PM_{10} concentration from each sensor (µg/m^3)
    b. FEM PM_{10} concentration (µg/m^3)
    c. FEM $PM_{2.5}$ concentration (µg/m^3)
    d. $PM_{2.5}/PM_{10}$ Ratio
    e. T (°C)
    f. RH (%)

Table 2-6. Elevated RH Test Conditions for PM_{10} Sensors

Parameter	Reference Setpoint
PM_{10} Concentration	60 µg/m^3 ± 5%
$PM_{2.5}/PM_{10}$ Ratio	≤ 0.4 ± 0.1
T	20°C ± 1°C
RH	85% ± 5%

2.2.5 Effect of Temperature (T)

To determine the effect of elevated T on sensor performance, take the steps below.

1. Repeat the procedure outlined in Section 2.2.3 of this report.
2. Supply the exposure chamber with the conditions in Table 2-7.
3. Allow the measurements to stabilize within the tolerances shown in Table 2-7.
4. Once steady state is achieved, collect either a minimum of 20-30 pairs of time-matched sensor and FEM data points or three (3) consecutive hours for the parameters listed below.
    a. PM_{10} concentration from each sensor (µg/m^3)
    b. FEM PM_{10} concentration (µg/m^3)
    c. FEM $PM_{2.5}$ concentration (µg/m^3)
    d. $PM_{2.5}/PM_{10}$ ratio
    e. T (°C)
    f. RH (%)

Table 2-7. Elevated T Test Conditions for PM_{10} Sensors

Parameter	Reference Setpoint
PM_{10} Concentration	60 µg/m^3 ± 5%
$PM_{2.5}/PM_{10}$ Ratio	≤ 0.4 ± 0.1
T	40°C ± 1°C
RH	40% ± 5%

2.2.6 Drift

See Figure 2-4 in the $PM_{2.5}$ Targets Report for a summary of the entire drift testing procedure.

2.2.6.1 Drift (Day 1) – Low and Mid Concentration Drift

To assess the drift, begin by testing at low and mid level PM_{10} concentrations to assess performance on Day 1. To do so, follow the steps below.

1. Supply the exposure chamber with the conditions shown in Table 2-8.
2. Allow the measurements to stabilize within the tolerances shown in Table 2-8.

3. Once steady state is achieved, collect either a minimum of 20-30 pairs of time-matched sensor and FEM data points or three (3) consecutive hours for the parameters listed below.
    a. PM_{10} concentration from each sensor (µg/m^3)
    b. FEM PM_{10} concentration (µg/m^3)
    c. FEM $PM_{2.5}$ concentration (µg/m^3)
    d. $PM_{2.5}/PM_{10}$ ratio
    e. T (°C)
    f. RH (%)
4. Supply the exposure chamber with the conditions shown in Table 2-9.
5. Allow the measurements to stabilize within the tolerances shown in Table 2-9.
6. Once steady state is achieved, collect either a minimum of 20-30 pairs of time-matched sensor and FEM data points or three (3) consecutive hours for the parameters listed below.
    a. PM_{10} concentration from each sensor (µg/m^3)
    b. FEM PM_{10} concentration (µg/m^3)
    c. FEM $PM_{2.5}$ concentration (µg/m^3)
    d. $PM_{2.5}/PM_{10}$ ratio
    e. T (°C)
    f. RH (%)

Table 2-8. Low Concentration Drift Test Conditions for PM_{10} Sensors

Parameter	Reference Setpoint
PM_{10} Concentration	15 µg/m^3 ± 10%
$PM_{2.5}/PM_{10}$ Ratio	≤ 0.4 ± 0.1
T	20°C ± 1°C
RH	40% ± 5%

Table 2-9. Mid Concentration Drift Test Conditions for PM_{10} Sensors

Parameter	Reference Setpoint
PM_{10} Concentration	60 µg/m^3 ± 5%
$PM_{2.5}/PM_{10}$ Ratio	≤ 0.4 ± 0.1
T	20°C ± 1°C
RH	40% ± 5%

2.2.6.2 Drift (Day 60) to Evaluate Sensor Aging

To assess sensor drift over a 60-day period, follow the steps below.

1. Operate the PM_{10} sensors in ambient, outdoor air for at least a consecutive 60-day period.

2. Following the 60-day period*, repeat the procedure in Section 2.2.6.1 in this report, with the aged sensors.

*The 60-day drift was chosen to balance the needs for a sufficient length of time in order to measure potential drift without the need to be unduly burdensome. It may be informative to repeat the drift test as sensors age providing additional data points at periodic intervals up to the expected lifespan of the sensor.

2.2.7 Accuracy at High Concentrations

To evaluate sensor accuracy at high PM_{10} concentrations, follow the steps below.

1. Supply the exposure chamber with the conditions in Table 2-10, with the high PM_{10} concentration of 200 μg/m^3.
2. Allow the measurements to stabilize within the tolerances shown in Table 2-10.
3. Once steady state is achieved, collect either a minimum of 20-30 pairs of time-matched sensor and FEM data points or three (3) consecutive hours for the parameters listed below.
    a. PM_{10} concentration from each sensor (μg/m^3)
    b. FEM PM_{10} concentration (μg/m^3)
    c. FEM $PM_{2.5}$ concentration (μg/m^3)
    d. $PM_{2.5}/PM_{10}$ ratio
    e. T (°C)
    f. RH (%)
4. Repeat Steps 1-3 with the higher PM_{10} concentration of 300 μg/m^3.

Table 2-10. High and Higher PM_{10} Concentration Test Conditions

Parameter	Reference Setpoint
PM_{10} Concentration (High)*	200 μg/m^3 ± 5%
PM_{10} Concentration (Higher)*	300 μg/m^3 ± 5%
$PM_{2.5}/PM_{10}$ Ratio	≤ 0.4 ± 0.1
T	20°C ± 1°C
RH	40% ± 5%

*PM_{10} concentrations can easily reach over 1000 μg/m^3 during dust events. However, some exposure chambers may have a challenge generating PM concentrations above 300 μg/m^3 and/or repeated exposure to high concentrations may require more extensive and frequent maintenance of the FEM. The high concentration value (200 μg/m^3) falls within the orange Air Quality Index (AQI) category (unhealthy for sensitive groups) and the higher concentration value (300 μg/m^3) falls with the red AQI category (unhealthy). Testing at the higher concentration value is optional but may be particularly useful for consumers interested in measurements during dust events.

3.0 Performance Metrics and Supporting Calculations for Evaluating PM_{10} Air Sensors

Testers should refer to Section 3.0 (*Performance Metrics and Supporting Calculations for Evaluating $PM_{2.5}$ Sensors*) in the $PM_{2.5}$ Targets Report for a general overview and other information about the recommended performance metrics.

Table 3-1 highlights the key similarities and differences for the performance metrics recommended for PM_{10} air sensors.

Table 3-1. Summary of Recommended Performance Metrics for $PM_{2.5}$ and PM_{10} Sensors (*gray boxes indicate differences*)

Test Type	Metric	$PM_{2.5}$	PM_{10}	Comments	Section Describing Calculation in $PM_{2.5}$ Targets Report
Base Testing	Precision	X	X	No change	3.1.3
	Bias	X	X	No change	3.1.4
	Linearity	X	X	No change	3.1.4
	Error	X	X	No change	3.1.5
	Exploring Meteorological Effects	X	X	No change	3.1.6
	$PM_{2.5}/PM_{10}$ Ratio		X	New metric for PM_{10} only	-
Enhanced Testing	Precision	X	X	No change	3.2.3
	Bias	X	X	No change	3.2.4
	Linearity	X	X	No change	3.2.4
	Error	X	X	No change	3.2.5
	Effects of Relative Humidity (RH)	X	X	No change	3.2.6
	Effects of Temperature (T)	X	X	No change	3.2.7
	Drift	X	X	No change	3.2.8
	Accuracy at High Concentrations	X	X	No change	3.2.9
	$PM_{2.5}/PM_{10}$ Ratio		X	New metric for PM_{10} only	-

3.1 Base Testing Calculations

Testers should review Section 3.1 (*Base Testing Calculations*) in the $PM_{2.5}$ Targets Report for important information on time aligning measurements and the FRM/FEM monitor operating schedules.

For all calculations in Sections 3.1.1 to 3.1.7 below, PM_{10} sensor, PM_{10} FRM/FEM, and/or $PM_{2.5}$ FRM/FEM data should be used.

3.1.1 Daily Averages

Follow guidance and Equation 1 (Eq. 1) in Section 3.1.1 (*Daily Averages*) in the $PM_{2.5}$ Targets Report to calculate daily (24-hour) averages for all instruments used during base testing.

3.1.2 Deployment Averages

Follow guidance and Eq. 2 in Section 3.1.2 (*Deployment Averages*) in the $PM_{2.5}$ Targets Report to calculate deployment averages for all instruments used during base testing.

3.1.3 Precision

Follow guidance in Section 3.1.3 (*Precision*) in the $PM_{2.5}$ Targets Report to calculate standard deviation (SD; see Eq. 3) and coefficient of variation (CV; see Eq. 4) for all instruments used during base testing.

3.1.4 Bias and Linearity

Follow guidance in Section 3.1.4 (*Bias and Linearity*) in the $PM_{2.5}$ Targets Report for all instruments used during base testing.

3.1.5 Error

Follow guidance in Section 3.1.5 (*Error*) in the $PM_{2.5}$ Targets Report to calculate the root mean square error (RMSE; see Eq. 5) and normalized root mean square error (NRMSE; see Eq. 6) for all instruments used during base testing.

3.1.6 Exploring Effect of Meteorology

Similar to $PM_{2.5}$ sensor technologies, research suggests that meteorology [specifically T, RH, and dew point (DP)] can also influence the performance of currently available PM_{10} sensor technologies (AQ-SPEC PM sensor evaluations; Jayaratne et al., 2018; Vogt et al., 2021).

Additionally, because air sensors are often low power and low flow, they may not efficiently aspirate coarse particles especially when flow must overcome the force of wind acting to move particles in another direction. The effects may be most pronounced when wind acts directly opposite the flow into the sensor.

3.1.6.1 Potential Scatter Plots

Follow guidance in Section 3.1.6.1 (*Potential Scatter Plots*) in the $PM_{2.5}$ Targets Report for all instruments used during base testing.

Testers may choose plots from the original list with the following additions:

- Concentration difference between the sensor and FRM/FEM $PM_{2.5}$ concentration vs. average wind speed

- Absolute concentration difference between the sensor and FRM/FEM $PM_{2.5}$ concentration as a function of wind direction

Because wind direction and speed can vary significantly throughout the day, these additional plots may be most informative at higher time resolution (e.g., 1-hour averages rather than 24-hour averages) if data are available.

3.1.6.2 Normalized Concentration

Follow guidance and Eq. 7 in Section 3.1.6.2 (*Normalized Concentration*) in the $PM_{2.5}$ Targets Report for all instruments used during base testing.

3.1.6.3 Concentration Difference and Absolute Concentration Difference

Follow guidance and Eq. 8a and Eq. 8b in Section 3.1.6.3 (*Concentration Difference and Absolute Concentration Difference*) in the $PM_{2.5}$ Targets Report for all instruments used during base testing.

3.1.6.4 Dew Point (DP)

Follow guidance and Eq. 9 in Section 3.1.6.4 [*Dew Point (DP)*] in the $PM_{2.5}$ Targets Report for all instruments used during base testing.

3.1.7 $PM_{2.5}/PM_{10}$ Ratio

The $PM_{2.5}/PM_{10}$ ratio can be used to understand the PM size distribution. This is particularly important for evaluating PM_{10} sensors as it can provide information on how well the sensor is measuring the PM_{10} size fraction. A $PM_{2.5}/PM_{10}$ should be calculated from valid 1-hour FRM/FEM data for each field test (Eq. 1S). This ratio can also be calculated using valid 1-hour sensor data for each field test.

$$PM_{2.5} \,/\, PM_{10} = \frac{R_{h,\,PM_{2.5}}}{R_{h,\,PM_{10}}} \qquad \text{Eq. 1S}$$

Where:

$PM_{2.5} \,/\, PM_{10}$ = $PM_{2.5}$ to PM_{10} ratio

$R_{h,\,PM_{2.5}}$ = valid 1-hour average FRM/FEM $PM_{2.5}$ concentration for hour *h* (µg/m^3)

$R_{h,\,PM_{10}}$ = valid 1-hour average FRM/FEM PM_{10} concentration for hour *h* (µg/m^3)

3.2 Enhanced Testing Calculations

Testers should review Section 3.2 (*Enhanced Testing Calculations*) in the $PM_{2.5}$ Targets Report for important information on time aligning measurements and data averaging.

For all calculations in Sections 3.2.1 to 3.2.10 below, PM_{10} sensor, PM_{10} FEM, and/or $PM_{2.5}$ FEM data should be used.

3.2.1 Data Averages

Follow guidance and Eq. 10 in Section 3.2.1 (*Data Averages*) in the $PM_{2.5}$ Targets Report to calculate data averages for all instruments used during enhanced testing.

3.2.2 Test Averages

Follow guidance and Eq. 11 in Section 3.2.2 (*Test Averages*) in the $PM_{2.5}$ Targets Report to calculate test averages for all instruments used during enhanced testing.

3.2.3 Precision

Follow guidance in Section 3.3.3 (*Precision*) in the $PM_{2.5}$ Targets Report to calculate standard deviation (SD; see Eq. 12) and coefficient of variation (CV; see Eq. 13) for all instruments used during enhanced testing.

3.2.4 Bias and Linearity

Follow guidance in Section 3.2.4 (*Bias and Linearity*) in the $PM_{2.5}$ Targets Report for all instruments used during enhanced testing.

3.2.5 Error

Follow guidance in Section 3.2.5 (*Error*) in the $PM_{2.5}$ Targets Report to calculate root mean square error (RMSE; see Eq. 14) and normalized root mean square error (NRMSE; see Eq. 15) for all instruments used during enhanced testing.

3.2.6 Effect of Relative Humidity (RH)

Follow guidance and Eq. 16 in Section 3.2.6 [*Effect of Relative Humidity (RH)*] in the $PM_{2.5}$ Targets Report for all instruments used during enhanced testing.

3.2.7 Effect of Temperature (T)

Follow guidance and Eq. 17 in Section 3.2.7 [*Effect of Temperature (T)*] in the $PM_{2.5}$ Targets Report for all instruments used during enhanced testing.

3.2.8 Drift

Follow guidance and Eq. 18 in Section 3.2.8 (*Drift*) in the $PM_{2.5}$ Targets Report for all instruments used during enhanced testing.

3.2.9 Accuracy at High Concentrations

Follow guidance and Eq. 19 in Section 3.2.9 (*Drift*) in the $PM_{2.5}$ Targets Report for all instruments used during enhanced testing.

3.2.10 $PM_{2.5}/PM_{10}$ Ratio

The $PM_{2.5}/PM_{10}$ ratio can be used to understand the PM size distribution. This is particularly important for evaluating PM_{10} sensors as it can provide information on how well the sensor is measuring the PM_{10} size fraction. A $PM_{2.5}/PM_{10}$ should be calculated from valid FEM data during the steady state test period (Eq. 2S). This ratio can also be calculated from valid sensor data during the steady state test period.

$$PM_{2.5} \,/\, PM_{10} = \frac{R_{t,\,PM_{2.5}}}{R_{t,\,PM_{10}}} \qquad \text{Eq. 2S}$$

Where:

$PM_{2.5} \,/\, PM_{10}$ = $PM_{2.5}$ to PM_{10} ratio

$R_{t,\,PM_{2.5}}$ = valid test average FEM $PM_{2.5}$ concentration for time *t* over the test period (μg/m^3)

$R_{t,\,PM_{10}}$ = valid test average FEM PM_{10} concentration for time *t* over the test period (μg/m^3)

4.0 Target Values for PM_{10} Air Sensors

4.1 Approach

Similar to the $PM_{2.5}$ Targets Report, the development of target values for the performance metrics for PM_{10} air sensors were informed by the following:

- Discussions during the 2019 Workshop on "Deliberating Performance Targets for Air Quality Sensors" (Duvall et al., 2021);
- Performance specifications for FEM monitors (40 CFR Part 53, Table C-4 to Subpart C);
- The U.S. EPA findings on air sensor evaluations (https://www.epa.gov/air-sensor-toolbox/evaluation-emerging-air-sensor-performance, *last accessed 04/25/2023*;
- South Coast AQMD Air Quality Sensor Performance Evaluation Center (AQ-SPEC) sensor field evaluations (http://www.aqmd.gov/aq-spec/evaluations/criteria-pollutants/summary-pm, *last accessed 10/04/2023*; South Coast AQMD, 2016); and
- Reviews of data quality levels published in peer-reviewed literature (U.S. EPA, 2018; U.S. EPA, 2020; and other references noted in Appendix D).

The sensor performance evaluation results gathered from the available resources are summarized in Table 4-1 (more detail available in Appendix D). In summarizing the performance results, the U.S. EPA did not consider results deemed to be outliers or unrepresentative of normal sensor operation to avoid significantly biasing the recommended target values. These results reflect out-of-box sensor performance before additional corrections were made by the user.

Table 4-1. PM_{10} Sensor Performance Field Evaluation Results from Available Resources

Performance Metric		Range		Average		Median	
		24-hour	1-hour	24-hour	1-hour	24-hour	1-hour
Precision	SD (µg/m^3)	4.40 to 26.80	0.82 to 26.10	13.38	13.73	11.50	14.40
Bias	Slope*	0.51 to 1.50	0.56 to 1.44	0.96	1.01	0.96	0.97
	Intercept* (µg/m^3)	-13.90 to 14.68	-12.16 to 9.41	-2.55	-3.73	-2.53	-4.09
Linearity	$R^{2,\#}$	0.50 to 0.98	0.76 to 0.96	0.86	0.74	0.86	0.74
Error	RMSE (µg/m^3)	2.70 to 42.50	5.60 to 38.70	21.30	20.97	21.63	19.95

Note: Resources include AQ-SPEC sensor evaluations, U.S. EPA sensor evaluations, and peer-reviewed literature. AQ-SPEC data, where the slope and intercept data were calculated by mathematical manipulation, was only considered if $R^2 \geq 0.75$

*Slopes outside of 0.5 to 1.5 were not considered; the intercept was not considered if the slope was discarded.

#R^2 values less than 0.5 were not considered.

4.2 List of Target Values

Table 4-2 summarizes the performance metrics and target values recommended for the base and enhanced testing protocols for PM_{10} air sensors used in ambient, outdoor, fixed site NSIM applications. Target values for $PM_{2.5}$ sensors are included for comparison. The recommended performance metrics and target values reflect the current state-of-the-science as the range of observed performance (Table 4-1) demonstrates that PM_{10} sensors should be able to achieve. Encouraging development of sensors which meet these target values should help ensure that sensor performance can be well characterized and understood.

Table 4-2. Base and Enhanced Testing – Recommended Performance Metrics and Target Values for $PM_{2.5}$ and PM_{10} Sensors Used in Ambient, Outdoor, Fixed Site NSIM Applications

Performance Metric		Target Value		
		$PM_{2.5}$ Base Testing	PM_{10} Base Testing	Enhanced Testing*
Precision	Standard Deviation (SD)	≤ 5 µg/m³	≤ 5 µg/m³	No target values recommended; report results
	-OR-			
	Coefficient of Variation (CV)	≤ 30%	≤ 30%	
Bias	Slope	1 ± 0.35	1 ± 0.35	
	Intercept (b)	-5 ≤ b ≤ 5 µg/m³	-10 ≤ b ≤ 10 µg/m³	
Linearity	Coefficient of Determination (R^2)	≥ 0.70	≥ 0.70	
Error	Root Mean Square Error (RMSE) or Normalized Root Mean Square Error (NRMSE)	RMSE ≤ 7 µg/m³ or NRMSE ≤ 30%†	RMSE ≤ 14 µg/m³ or NRMSE ≤ 30%†	

*Applies to both PM_{10} and $PM_{2.5}$ sensors
†A sensor will meet this target if either the RMSE or NRMSE meet this criterion. See Appendix D for further discussion.

Additional performance metrics and test conditions for the enhanced testing protocol are shown in Table 4-3. Information for $PM_{2.5}$ sensors is also included for comparison. Target values for enhanced testing are not included at this time due to limited feasibility, lack of consistency regarding testing protocols, and inconsistency in sensor evaluation results that can result due to the limited amount of data that will be collected and the variation in the tester's choice of PM surrogate (see Appendix D for more detailed discussion).

.

Table 4-3. Enhanced Testing – Additional Recommended Performance Metrics and Test Conditions for $PM_{2.5}$ and PM_{10} Sensors Used in Ambient, Outdoor, Fixed Site NSIM Applications

Performance Metric	Test Conditions			Section Describing Calculation in this Report
	Parameter	$PM_{2.5}$	PM_{10}	
Effect of Relative Humidity (RH)	Moderate RH	40% ± 5%	40% ± 5%	3.2.6
	Elevated RH	85% ± 5%	85% ± 5%	3.2.6
Effect of Temperature (T)	Moderate T	20°C ± 1°C	20°C ± 1°C	3.2.7
	Elevated T	40°C ± 1°C	40°C ± 1°C	3.2.7
Drift	Low concentration	10 µg/m^3 ± 10%	15 µg/m^3 ± 10%	3.2.8
	Mid concentration	35 µg/m^3 ± 5%	60 µg/m^3 ± 5%	3.2.8
Accuracy at High Concentrations	High concentration	150 µg/m^3 ± 5%	200 µg/m^3 ± 5%	3.2.9
	Higher concentration	250 µg/m^3 ± 5%	300 µg/m^3 ± 5%	3.2.9

It is recognized that the information in this report is based on the current knowledge of PM_{10} air sensors at the time this report was released and that PM_{10} sensor technologies will likely continue to develop and improve over time. The U.S. EPA anticipates updating Tables 4-2 and 4-3 as well as other information in this report, as feasible, to reflect advances in PM_{10} sensor technologies and knowledge gained from sensor evaluation results. Updates will likely be shared as an addendum to this report.

Testing results do not constitute certification or endorsement by the U.S. EPA. It is recommended that testers make the testing reports available on their respective websites to inform consumers on the testing results.

5.0 References

1. Castell, N., Dauge, F.R., Schneider, P., Vogt, M., Lerner, U., Fishbain, B., Broday, D., Bartonova, A. Can commercial low-cost sensor platforms contribute to air quality monitoring and exposure estimates? Environment International, 99, 293-302, http://dx.doi.org/10.1016/j.envint.2016.12.007, 2017.
2. Crilley, L.R., Shaw, M., Pound, R., Kramer, L.J., Price, R., Young, S., Lewis, A.C., Pope, F.D. Evaluation of a low-cost optical particle counter (Alphasense OPC-N2) for ambient air monitoring. Atmospheric Measurement Techniques, 11, 709-720, https://doi.org/10.5194/amt-11-709-2018, 2018.
3. Duvall, R.M., Hagler, G.S.W., Clements, A.L., Benedict, K., Barkjohn, K., Kilaru, V., Hanley, T., Watkins, N., Kaufman, A., Kamal, A., Reece, S., Fransioli, P., Gerboles, M., Gillerman, G., Habre, R., Hannigan, M., Ning, Z., Papapostolou, V., Pope, R., Quintana, P.J.E., Lam Snyder, J. Deliberating performance targets: Follow-on workshop discussing PM_{10}, NO_2, CO, and SO_2 air sensor targets. Atmospheric Environment: X, 246, https://doi.org/10.1016/j.atmosenv.2020.118099, 2021.
4. Feinberg, S., Williams, R., Hagler, G.S.W., Rickard, J., Brown, R., Garver, D., Harshfield, G., Stauffer, P., Mattson, E., Judge, R., Garvey, S. Long-term evaluation of air sensor technology under ambient conditions in Denver, Colorado. Atmospheric Measurement Techniques, 11, 4605-4615, https://doi.org/10.5194/amt-11-4605-2018, 2018.
5. Jayaratne, R., Liu, X., Thai, P., Dunbabin, M., Morawska, L. The influence of humidity on the performance of low-cost air particle mass sensor and the effect of atmospheric fog. Atmospheric Measurement Techniques, 11, 4883-4890, https://doi.org/10.5194/amt-11-4883-2018, 2018.
6. Kaur, K., Kelly, K.E. Performance evaluation of the Alphasense OPC-N3, and the Plantower PMS5003 sensor in measuring dust events in Salt Lake Valley, Utah. Atmospheric Measurement Techniques, 16, 2455-2470, https://doi.org/10.5194/amt-16-2455-2023, 2023.
7. Kulkarni, P., Baron, P.A., Willeke, K. Aerosol Measurement: Principles, Techniques, and Applications. 3rd Edition, John Wiley & Sons, Inc., 2011.
8. Mukherjee, A., Stanton, L.G., Graham, A.R., Roberts, P.T. Assessing the utility of low-cost particulate matter sensors over a 12-week period in the Cuyama Valley of California. Sensors, 17, 1805, http://dx.doi.org/10.3390/s17081805, 2017.
9. Papapostolou, V., Zhang, H., Feenstra, B.J., Polidori, A. Development of an environmental chamber for evaluating the performance of low-cost air quality sensors under controlled conditions. Atmospheric Environment, 171, 82-90, https://doi.org/10.1016/j.atmosenv.2017.10.003, 2017.
10. Qin, X., Hou, L., Gao, J., Si, S. The evaluation and optimization of calibration methods for low-cost particulate matter sensors: Inter-comparison between fixed and mobile methods. Science of The Total Environment, 715, 136791, https://doi.org/10.1016/j.scitotenv.2020.136791, 2020.
11. Rogulski, M., Badyda, A. Investigation of low-cost and optical particulate matter sensors for ambient monitoring. Atmosphere, 11, 1040, https://www.mdpi.com/2073-4433/11/10/1040, 2020.

12. Shahsavani, A., Naddafi, K., Jafarzade Haghighifard, N., Mesdaghinia A., Yunesian, M., Nabizadeh, R., Arahami, M., Sowlat, M.H., Yarahmadi, M., Saki, H., Alimohamadi, M., Nazmara, S., Motevalian, S.A., Goudarzi, G. The evaluation of PM_{10}, $PM_{2.5}$, and PM1 concentrations during Middle Eastern Dust (MED) events in Ahvaz, Iran, from April through September 2010. Journal of Arid Environments, 77, 72-83, http://dx.doi.org/10.1016/j.jaridenv.2011.09.007, 2012.
13. South Coast AQMD, Air Quality Sensor Performance Evaluation Center (AQ-SPEC). PM sensor evaluations. Available at: http://www.aqmd.gov/aq-spec/evaluations/summary-pm.
14. South Coast AQMD, Air Quality Sensor Performance Evaluation Center (AQ-SPEC). Laboratory Evaluation of Low-Cost Air Quality Sensors: Laboratory Setup and Testing Protocol, 2016. Available at: http://www.aqmd.gov/docs/default-source/aq-spec/protocols/sensors-lab-testing-protocol6087afefc2b66f27bf6fff00004a91a9.pdf.
15. Sugimoto, N., Shimizu, A., Matsui, I., Nishikawa, M. A method for estimating the fraction of mineral dust in particulate matter using $PM_{2.5}$-to-PM_{10} ratios. Particuology, 28, 114-120, 699, https://doi.org/10.1016/j.partic.2015.09.005, 2016.
16. Tagle, M., Rojas, F., Reyes, F., Vásquez, Y., Hallgren, F., Lindén, J., Kolev, D., Watne, A.K., Oyola, P. Field performance of a low-cost sensor in the monitoring of particulate matter in Santiago, Chile. Environmental Monitoring and Assessment 192:171, https://link.springer.com/article/10.1007/s10661-020-8118-4, 2020.
17. U.S. EPA. Quality Assurance Handbook for Air Pollution Measurement Systems Volume II – Ambient Air Quality Monitoring Program, U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards, EPA-454/B-17-001, 2017.
18. U.S. EPA. Peer Review and Supporting Literature Review of Air Sensor Technology Performance Targets. U.S. Environmental Protection Agency, Office of Research and Development, Research Triangle Park, NC, EPA/600/R-18/324, 2018.
19. U.S. EPA. Peer Review and Supporting Literature Review of Air Sensor Technology Performance Targets: 2019 Supplemental. U.S. Environmental Protection Agency, Office of Research and Development, Research Triangle Park, NC, EPA/600/R-20/120, 2020.
20. U.S. EPA. Performance Testing Protocols, Metrics, and Target Values for Fine Particulate Matter Air Sensors: Use in Ambient, Outdoor, Fixed Site, Non-regulatory Supplemental and Informational Monitoring Applications. U.S. Environmental Protection Agency, Office of Research and Development, Research Triangle Park, NC, EPA/600/R-20/280, 2021.
21. U.S. EPA, 40 CFR Part 50 – National Primary and Secondary Ambient Air Quality Standards.
22. U.S. EPA, 40 CFR Part 53 – Ambient Air Monitoring Reference and Equivalent Methods.
23. U.S. EPA 40 CFR Part 58 – Ambient Air Quality Surveillance.
24. Vincent, J.H. Aerosol Sampling: Science, Standards, Instrumentation and Applications. 1st Edition, John Wiley & Sons, 2007.
25. Vogt, M., Schneider, P., Castell, N., Hamer, P. Assessment of low-cost particulate matter sensor systems against optical and gravimetric methods in a field co-location in Norway. Atmosphere, 12, 961, https://www.mdpi.com/2073-4433/12/8/961, 2021.

Appendix A: Definitions

This Appendix summarizes definitions for new terms included throughout this report as well as important terms that were not included in the $PM_{2.5}$ Targets Report. In developing these definitions, we consulted a variety of resources (e.g., AQ-SPEC, 40 CFR Part 53 and 58, peer-reviewed literature) to try to provide consistency in the use of these terms among documents and an appropriate level of detail to support testers and consumers. For definitions of the other commonly used terms in this supplemental report, please refer to Appendix A (*Definitions*) in the $PM_{2.5}$ Targets Report.

Aspiration: A process where a known volume of air containing aerosol particles is sampled by pulling or sucking it with the aid of pump through one or more openings in an enclosure (Vincent, 2007).

Aspiration Efficiency: The fraction of particles entering an inlet from the ambient environment (Kulkarni et al., 2011).

Monitoring Spatial Scale: In siting monitoring stations, the goal is to correctly match the spatial scale represented by the sample of monitored air with the spatial scale most appropriate for the monitoring objective of the station. The representative measurement scales of greatest interest are: 1) Microscale – concentrations in air volumes associated with area dimensions ranging from several meters up to about 100 meters, 2) Middle scale – concentrations typical of areas up to several city blocks in size with dimensions ranging from about 100 meters to 0.5 kilometers, 3) Neighborhood scale – concentrations within some extended area of the city that has relatively uniform land use with dimensions in the 0.5 to 4.0 kilometer range, 4) Urban scale – overall, citywide conditions with dimensions on the order of 4 to 50 kilometers; this scale would usually require more than one site for definition, 5) Regional scale – usually a rural area of reasonably homogenous geography and extends from tens to hundreds of kilometers, and 6) National and global scales – concentrations characterizing the nation and the globe as a whole (U.S. EPA, 2017).

$PM_{2.5}$: Also referred to as fine particulate matter (PM), inhalable particles with diameters that are generally less than 2.5 µm.

PM_{10}: Inhalable particles with diameters that are generally less than 10 µm. PM_{10} includes both the coarse and fine particulate matter ($PM_{2.5}$) size fractions.

$PM_{10-2.5}$: Also referred to as coarse particles or coarse particulate matter (PMc), inhalable particles with diameters that are generally larger than 2.5 µm and smaller than, or equal to,10 µm in diameter.

$PM_{2.5}/PM_{10}$ Ratio: Provides information on the size distribution of particulate matter and how particles are formed in the atmosphere. Smaller ratios suggest that larger particles are more dominate, whereas larger ratios indicate that smaller particles are more dominate.

Appendix B: Supporting Information for Testing Protocols

Testing protocols for PM_{10} air sensors were drafted based on known best practices in the literature to date. This includes research performed by AQ-SPEC (Papapostolou et al., 2017; South Coast AQMD, 2016), the U.S. EPA, and the People's Republic of China MEE, as well as protocols used for FRM/FEM regulatory monitors (40 CFR Part 53). The goal of providing these test protocols is to collect an array of comparable data on air sensors without overstraining resources.

Base Testing Protocols were modeled after field evaluations conducted by a variety of organizations with slight variations. Sensor studies most commonly compare sensor measurements at 24-hour or 1-hour averages. This protocol emphasizes 24-hour averaged data to compare air sensor and FRM/FEM monitor information as there is little variation between different types of FRM/FEM monitors at this scale per federal requirements. Thus, with the 24-hour averaged data there is more confidence that the type of FRM/FEM monitor will not influence the results. If the selected FEM monitor can provide 1-hour averaged data, testers are encouraged to develop a second testing report using 1-hour average data as it can be reasonably assumed that this averaging interval will be used in practice by consumers. Testers and consumers should be aware that there may be more variation in performance among the various makes/models of FEM monitors at this time averaging interval. The base testing protocol uses a single FRM/FEM monitor for PM_{10} and $PM_{2.5}$ as the recommended minimum for comparison to reduce equipment needed for testing and to simplify calculations. In concurrence with current sensor evaluation efforts and 40 CFR Part 53.34, these testing protocols recommend testing three (3) or more sensors simultaneously for at least 30 days with a 75% data completeness threshold. Testing three (3) or more identical sensors can help consumers understand the variation in performance that may exist among identical sensors. Discussion with experts and review of current practices determined that two (2) deployments are likely sufficient to show sensor performance over a range of conditions including T, RH, weather, PM_{10} and $PM_{2.5}$ concentrations, and other factors that provide information about the sensor's potential performance in different areas of the U.S provided that there are meaningful variations in PMc compared to $PM_{2.5}$. Testing locations that meet the test site criteria will likely experience the necessary variations for effectively evaluating PM_{10} sensors but may not be as widespread, therefore testing in adjacent U.S. climate regions is appropriate. Two locations impacted by different sources (e.g., windblown dust, industrial coarse PM) would be most informative for testing.

Identification of field sites for base testing involved evaluating historical data from the Air Quality System (AQS) and other sources. Three years (2017 to 2019) of 24-hour averaged PM_{10} data from over 400 regulatory air monitoring sites [AQS, Interagency Monitoring of Protected Visual Environments (IMPROVE), and National Air Pollution Surveillance (NAPS, Canada)] were evaluated. Daily measurements were averaged to monthly intervals. Figure B-1 summarizes average PM_{10} concentrations for one month representing each season (winter, spring, summer, and fall) across the continental U.S. These maps provide a general overview of locations and seasons in which higher PM_{10}

concentrations may be found. While not all months are displayed, it can be reasonably assumed that the PM_{10} concentrations are likely similar for other months during each season.

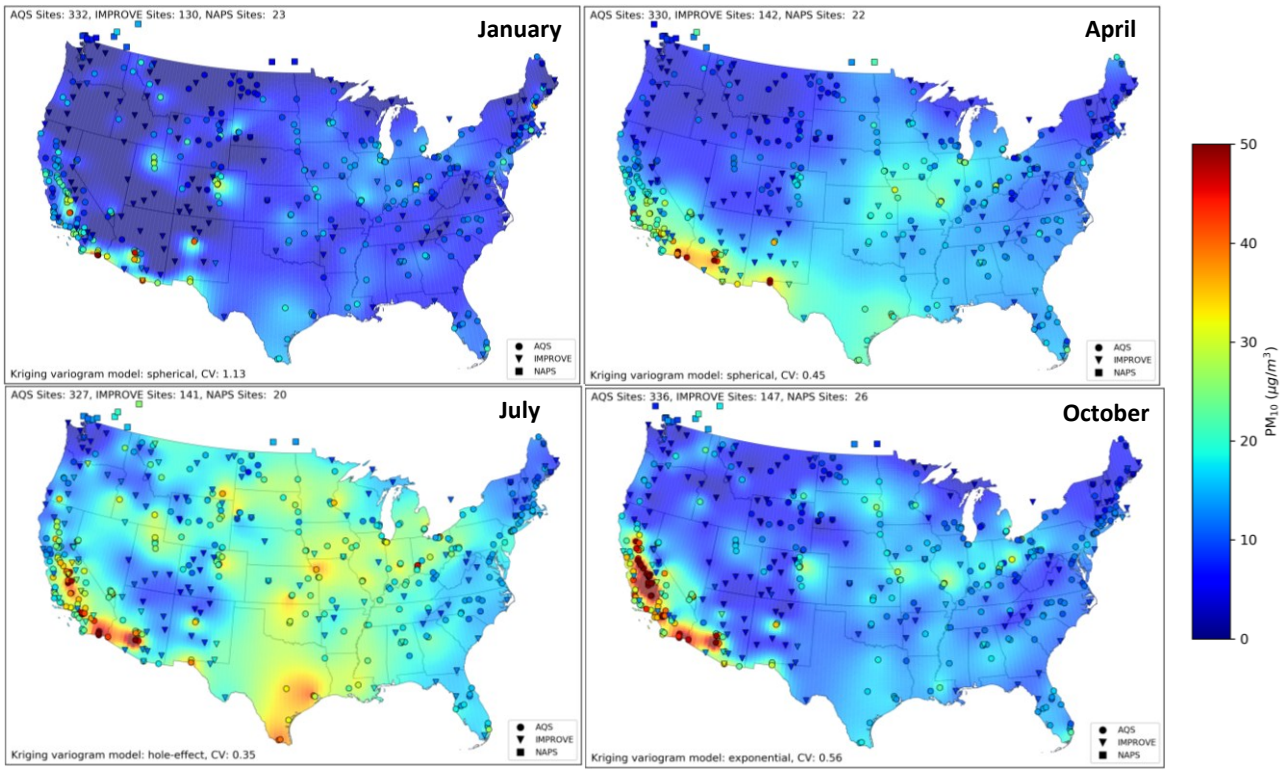


Figure B-1. Average PM_{10} Concentrations for Select Months over 2017-2019 across the Continental U.S.

To further understand PM_{10} concentrations and $PM_{2.5}/PM_{10}$ ratios across the U.S., 5 years (2017 to 2021) of PM_{10} and $PM_{2.5}$ data were obtained from 23 AQS sites. These sites were selected as they have collocated $PM_{2.5}$ and PM_{10} FRM/FEM monitors, and are anticipated to be impacted, at minimum periodically, by significant PM_{10} sources (e.g., windblown dust, industrial coarse PM). Analysis of the data showed that a minimum target 24-hour average PM_{10} concentration of 40 μg/m^3 was reached at least once for three or more months throughout the year at all the sites evaluated. A goal concentration of 40 μg/m^3 for one day during the 30-day test period was chosen as it could most likely be achieved at many sites and would provide flexibility in site selection. This value is larger than that recommended in the $PM_{2.5}$ Targets Report (25 μg/m^3) due to higher ambient PM_{10} concentrations, on average, than $PM_{2.5}$. Similar to the $PM_{2.5}$ Targets Report, no seasonal recommendation is provided for PM_{10} as it does not seem to be needed. Note that some historically high PM_{10} concentrations may be caused by extreme events that may not be present every year (e.g., dust storms, wildfires, construction activities) while others are caused by common seasonal events and typical meteorology.

$PM_{2.5}/PM_{10}$ ratios (also referred to as PM ratios) were investigated as an easily calculable metric for determining if a meaningful concentration of PMc is present and therefore likely also meaningful variations in PMc compared to $PM_{2.5}$. Variations in PMc that are different

from $PM_{2.5}$ variations will help to determine if the sensor can indeed see larger particles. To determine an appropriate target for choosing a testing location, PM ratios were calculated using historical data collected at regulatory monitoring sites in the West, Southwest, and Southeast by dividing the $PM_{2.5}$ concentration by the PM_{10} concentration at several time averaging intervals (e.g., 1-hour, 3-hour, 6-hour, 24-hour). Seven different PM ratio criteria were considered: ≤ 0.1, ≤ 0.2, ≤ 0.3, ≥ 0.4, ≥ 0.5, ≥ 0.6, and ≥ 0.7. Lower PM ratios indicate that larger particles are more dominate, while higher PM ratios indicate that smaller particles are more dominate. PM ratios between 0.1 to 0.3 have been observed during dust storm events (Kaur and Kelly, 2023; Sugimoto et al., 2016; Shahsavani et al., 2012). The percentage of days when the PM ratio satisfied each criterion for each time averaging interval was calculated. The analysis showed that it is more difficult to achieve a 24-hour goal PM ratio especially at ratios below 0.3 but a 1-hour PM ratio goal < 0.3 appeared more achievable. Further, $PM_{10-2.5}$ concentrations are expected to be more variable over shorter time periods so a wider range in the PM ratio will likely be observed over 1-hour. A goal $PM_{2.5}/PM_{10}$ ratio of ≤ 0.4 ± 0.1 for at least 1-hour during the 30-day test period was chosen for the test site criteria and it was observed in approximately 68% of the sites evaluated, captures conditions observed during dust storms, and would provide more flexibility for testers. Ideally this goal would be achieved for more than a single hour during the test period, however a minimum of 1-hour could provide adequate information for evaluating PM_{10} sensor performance.

Tables B-1 to B-4 summarize the AQS sites evaluated across the U.S. and seasons that are likely to meet the test site selection criteria (shown in Table 2-2).

Table B-1. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Winter Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} concentrations are ≥ 40 μg/m^3 at least once.

U.S. Climate Region	State	AQS Site ID	Monitoring Spatial Scale*	$PM_{2.5}/PM_{10}$ Ratio [median (min, max)]		
				December	January	February
West	California	06-019-5001	Neighborhood	0.65 (0, 1.0)	0.68 (0, 1.0)	0.50 (0, 1.0)
		06-029-0018	Neighborhood	0.29 (0, 1.0)	0.29 (0, 1.0)	0.29 (0, 1.0)
		06-037-9033	Neighborhood	0.32 (0, 1.0)	0.33 (0, 1.0)	0.31 (0, 1.0)
	Nevada	32-003-0043	Neighborhood	0.35 (0.06, 0.83)	0.35 (0, 1.0)	0.31 (0, 1.0)
		32-003-0298	Middle	0.36 (0.07, 0.80)	0.37 (0, 1.0)	0.37 (0, 1.0)
		32-003-0540	Neighborhood	0.38 (0, 0.97)	0.37 (0, 1.0)	0.30 (0, 1.0)
		32-003-0561	Neighborhood	0.38 (0.07, 1.0)	0.38 (0.01, 1.0)	0.30 (0.01, 1.0)
		32-003-1019	Regional	0.24 (0.05, 0.77)	0.26 (0, 1.0)	0.24 (0, 1.0)
Southwest	Arizona	04-013-0019	Neighborhood	0.44 (0, 1.0)	0.45 (0, 1.0)	0.38 (0, 1.0)
		04-013-1004	Neighborhood	0.51 (0, 1.0)	0.54 (0, 1.0)	0.45 (0, 1.0)
		04-013-4003	Neighborhood	0.47 (0, 1.0)	0.44 (0, 1.0)	0.37 (0, 1.0)
		04-013-9812	Neighborhood	0.35 (0, 1.0)	0.34 (0, 1.0)	0.28 (0, 1.0)
		04-013-9997	Neighborhood	0.38 (0, 1.0)	0.36 (0, 1.0)	0.29 (0, 1.0)
		04-021-3015	Middle	0.20 (0, 1.0)	0.22 (0, 1.0)	0.18 (0, 1.0)
	New Mexico	35-001-0023	Neighborhood	0.36 (0, 0.97)	0.38 (0, 1.0)	0.33 (0, 1.0)
		35-001-0029	Neighborhood	0.28 (0.04, 0.71)	0.30 (0.04, 1.0)	0.23 (0.05, 0.77)
		35-013-0016	Neighborhood	0.22 (0, 1.0)	0.23 (0, 1.0)	0.21 (0, 1.0)
Southeast	Alabama	01-073-0023	Neighborhood	0.57 (0, 0.98)	0.56 (0.10, 1.0)	0.59 (0.19, 1.0)
	Florida	12-095-2002	Neighborhood	0.53 (0, 1.0)	0.47 (0, 1.0)	0.47 (0, 1.0)
	Georgia	13-089-0002	Neighborhood	0.63 (0.19, 0.95)	0.63 (0, 1.0)	0.61 (0, 1.0)
		13-245-0091	Neighborhood	0.71 (0.12, 1.0)	0.77 (0.23, 1.0)	0.79 (0.17, 1.0)
	North Carolina	37-063-0015	Neighborhood	0.67 (0, 1.0)	0.69 (0, 1.0)	0.69 (0, 1.0)
		37-183-0014	Neighborhood	0.68 (0, 1.0)	0.68 (0, 1.0)	0.67 (0, 1.0)

*Neighborhood = 0.5 to 4 kilometers, Middle = 100 to 500 meters, Regional = tens to hundreds kilometers

Table B-2. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Spring Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} concentrations are ≥ 40 μg/m^3 at least once.

U.S. Climate Region	State	AQS Site ID	Monitoring Spatial Scale*	$PM_{2.5}/PM_{10}$ Ratio [median (min, max)]		
				March	April	May
West	California	06-019-5001	Neighborhood	0.43 (0, 1.0)	0.37 (0, 1.0)	0.24 (0, 1.0)
		06-029-0018	Neighborhood	0.33 (0, 1.0)	0.35 (0, 1.0)	0.32 (0, 1.0)
		06-037-9033	Neighborhood	0.33 (0, 1.0)	0.29 (0, 1.0)	0.30 (0, 1.0)
	Nevada	32-003-0043	Neighborhood	0.29 (0, 1.0)	0.26 (0, 0.98)	0.29 (0.11, 0.69)
		32-003-0298	Middle	0.30 (0, 0.96)	0.26 (0, 1.0)	0.3 (0.11, 0.73)
		32-003-0540	Neighborhood	0.27 (0, 1.0)	0.23 (0, 0.92)	0.32 (0.09, 0.71)
		32-003-0561	Neighborhood	0.27 (0, 0.99)	0.24 (0, 1.0)	0.31 (0.10, 0.68)
		32-003-1019	Regional	0.29 (0, 1.0)	0.26 (0, 1.0)	0.22 (0.09, 0.65)
Southwest	Arizona	04-013-0019	Neighborhood	0.31 (0, 1.0)	0.26 (0, 0.82)	0.26 (0, 0.74)
		04-013-1004	Neighborhood	0.40 (0, 1.0)	0.33 (0, 1.0)	0.32 (0, 0.90)
		04-013-4003	Neighborhood	0.30 (0, 1.0)	0.26 (0, 0.87)	0.22 (0, 0.81)
		04-013-9812	Neighborhood	0.26 (0, 1.0)	0.22 (0, 0.82)	0.22 (0, 0.80)
		04-013-9997	Neighborhood	0.25 (0, 1.0)	0.22 (0, 1.0)	0.22 (0, 1.0)
		04-021-3015	Middle	0.18 (0, 1.0)	0.18 (0, 1.0)	0.19 (0, 1.0)
	New Mexico	35-001-0023	Neighborhood	0.28 (0, 1.0)	0.25 (0, 1.0)	0.23 (0, 1.0)
		35-001-0029	Neighborhood	0.21 (0.04, 0.67)	0.20 (0.06, 0.73)	0.18 (0.05, 0.62)
		35-013-0016	Neighborhood	0.18 (0, 1.0)	0.18 (0, 1.0)	0.19 (0, 1.0)
Southeast	Alabama	01-073-0023	Neighborhood	0.53 (0, 0.99)	0.49 (0, 1.0)	0.51 (0, 1.0)
	Florida	12-095-2002	Neighborhood	0.47 (0, 1.0)	0.47 (0, 1.0)	0.44 (0, 1.0)
	Georgia	13-089-0002	Neighborhood	0.58 (0, 1.0)	0.51 (0, 1.0)	0.47 (0, 1.0)
		13-245-0091	Neighborhood	0.78 (0.25, 1.0)	0.74 (0.09, 1.0)	0.74 (0.16, 1.0)
	North Carolina	37-063-0015	Neighborhood	0.67 (0, 1.0)	0.58 (0, 1.0)	0.56 (0, 1.0)
		37-183-0014	Neighborhood	0.63 (0, 1.0)	0.55 (0, 1.0)	0.56 (0, 1.0)

*Neighborhood = 0.5 to 4 kilometers; Middle = 100 to 500 meters; Regional = tens to hundreds kilometers

Table B-3. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Summer Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} Concentrations are ≥ 40 µg/m^3 at least once.

U.S. Climate Region	State	AQS Site ID	Monitoring Spatial Scale*	$PM_{2.5}/PM_{10}$ Ratio [median (min, max)]		
				June	July	August
West	California	06-019-5001	Neighborhood	0.23 (0, 0.96)	0.26 (0, 1.0)	0.28 (0, 0.98)
		06-029-0018	Neighborhood	0.28 (0, 1.0)	0.33 (0, 1.0))	0.33 (0, 1.0)
		06-037-9033	Neighborhood	0.30 (0, 1.0)	0.31 (0, 1.0)	0.31 (0, 1.0)
	Nevada	32-003-0043	Neighborhood	0.23 (0, 1.0)	0.27 (0.08, 0.60)	0.29 (0.09, 0.69)
		32-003-0298	Middle	0.23 (0, 0.98)	0.27 (0, 0.98)	0.30 (0, 0.99)
		32-003-0540	Neighborhood	0.22 (0, 1.0)	0.26 (0, 1.0)	0.27 (0, 0.99)
		32-003-0561	Neighborhood	0.23 (0, 0.94)	0.26 (0, 0.99)	0.27 (0, 0.97)
		32-003-1019	Regional	0.23 (0, 0.99)	0.27 (0.05, 0.62)	0.29 (0.07, 0.69)
Southwest	Arizona	04-013-0019	Neighborhood	0.27 (0, 0.97)	0.32 (0, 0.94)	0.32 (0, 0.98)
		04-013-1004	Neighborhood	0.31 (0, 0.81)	0.35 (0, 1.0)	0.38 (0.02, 1.0)
		04-013-4003	Neighborhood	0.21 (0, 0.76)	0.25 (0, 1.0)	0.27 (0, 0.98)
		04-013-9812	Neighborhood	0.22 (0, 0.85)	0.24 (0, 1.0)	0.26 (0, 0.96)
		04-013-9997	Neighborhood	0.23 (0, 1.0)	0.25 (0, 1.0)	0.25 (0, 1.0)
		04-021-3015	Middle	0.21 (0, 1.0)	0.21 (0, 1.0)	0.21 (0, 1.0)
	New Mexico	35-001-0023	Neighborhood	0.28 (0, 0.94)	0.32 (0, 0.97)	0.32 (0, 1.0)
		35-001-0029	Neighborhood	0.22 (0.05, 0.75)	0.26 (0.04, 0.61)	0.26 (0.04, 0.61)
		35-013-0016	Neighborhood	0.25 (0, 1.0)	0.27 (0, 1.0)	0.27 (0, 1.0)
Southeast	Alabama	01-073-0023	Neighborhood	0.48 (0, 1.0)	0.47 (0, 1.0)	0.51 (0, 1.0)
	Florida	12-095-2002	Neighborhood	0.45 (0, 1.0)	0.49 (0, 1.0)	0.47 (0, 1.0)
	Georgia	13-089-0002	Neighborhood	0.45 (0, 1.0)	0.46 (0.19, 0.81)	0.47 (0.20, 0.72)
		13-245-0091	Neighborhood	0.45 (0, 1.0)	0.68 (0.27, 1.0)	0.66 (0, 1.0)
	North Carolina	37-063-0015	Neighborhood	0.60 (0, 1.0)	0.67 (0, 1.0)	0.60 (0, 1.0)
		37-183-0014	Neighborhood	0.58 (0, 1.0)	0.60 (0, 1.0)	0.57 (0, 1.0)

*Neighborhood = 0.5 to 4 kilometers; Middle = 100 to 500 meters; Regional = tens to hundreds kilometers

Table B-4. 1-hour Average $PM_{2.5}/PM_{10}$ Ratios During Fall Months at Select AQS Sites (from 2017-2021). Gray shaded cells indicate that the 24-hour averaged PM_{10} Concentrations are ≥ 40 μg/m^3 at least once.

U.S. Climate Region	State	AQS Site ID	Monitoring Spatial Scale*	$PM_{2.5}/PM_{10}$ Ratio [median (min, max)]		
				September	October	November
West	California	06-019-5001	Neighborhood	0.23 (0, 1.0)	0.27 (0, 1.0)	0.50 (0, 1.0)
		06-029-0018	Neighborhood	0.33 (0, 1.0)	0.33 (0, 1.0)	0.23 (0, 1.0)
		06-037-9033	Neighborhood	0.28 (0, 1.0)	0.23 (0, 1.0)	0.24 (0, 1.0)
	Nevada	32-003-0043	Neighborhood	0.27 (0.08, 0.70)	0.26 (0.06, 0.58)	0.28 (0.06, 0.79)
		32-003-0298	Middle	0.26 (0, 1.0)	0.25 (0.06, 0.73)	0.28 (0, 0.75)
		32-003-0540	Neighborhood	0.24 (0, 1.0)	0.25 (0, 0.98)	0.28 (0.01, 0.97)
		32-003-0561	Neighborhood	0.25 (0, 0.99)	0.25 (0.06, 0.57)	0.30 (0.07, 0.68)
		32-003-1019	Regional	0.26 (0.05, 0.73)	0.23 (0.06, 0.58)	0.21 (0, 0.81)
Southwest	Arizona	04-013-0019	Neighborhood	0.29 (0, 1.0)	0.26 (0, 1.0)	0.28 (0, 1.0)
		04-013-1004	Neighborhood	0.34 (0, 1.0)	0.30 (0, 1.0)	0.32 (0, 1.0)
		04-013-4003	Neighborhood	0.25 (0, 0.97)	0.23 (0, 1.0)	0.27 (0, 1.0)
		04-013-9812	Neighborhood	0.25 (0, 1.0)	0.22 (0, 1.0)	0.23 (0, 1.0)
		04-013-9997	Neighborhood	0.26 (0, 1.0)	0.23 (0, 1.0)	0.27 (0, 1.0)
		04-021-3015	Middle	0.19 (0, 1.0)	0.15 (0, 1.0)	0.16 (0, 1.0)
	New Mexico	35-001-0023	Neighborhood	0.31 (0, 1.0)	0.24 (0, 1.0)	0.29 (0, 0.92)
		35-001-0029	Neighborhood	0.25 (0.05, 0.64)	0.20 (0.04, 0.69)	0.22 (0.05, 0.72)
		35-013-0016	Neighborhood	0.24 (0, 1.0)	0.20 (0, 1.0)	0.20 (0, 1.0)
Southeast	Alabama	01-073-0023	Neighborhood	0.49 (0, 1.0)	0.50 (0, 1.0)	0.56 (0, 1.0)
	Florida	12-095-2002	Neighborhood	0.47 (0, 1.0)	0.44 (0, 1.0)	0.48 (0, 1.0)
	Georgia	13-089-0002	Neighborhood	0.50 (0.11, 0.74)	0.49 (0.17, 0.77)	0.60 (0.14, 0.87)
		13-245-0091	Neighborhood	0.63 (0.25, 1.0)	0.64 (0.12, 1.0)	0.71 (0.17, 1.0)
	North Carolina	37-063-0015	Neighborhood	0.50 (0, 1.0)	0.50 (0, 1.0)	0.67 (0, 1.0)
		37-183-0014	Neighborhood	0.56 (0, 1.0)	0.55 (0, 1.0)	0.65 (0, 1.0)

*Neighborhood = 0.5 to 4 kilometers; Middle = 100 to 500 meters; Regional = tens to hundreds kilometers

Enhanced testing protocols were modeled after laboratory evaluations conducted by a variety of organizations seeking to quantify the effects of T and RH, drift, and accuracy at high concentrations. The PM_{10} concentration levels used in testing slightly exceed the Air Quality Index (AQI) breakpoints for 24-hour averaged PM_{10} concentrations.

Testing duration and data time averaging recommendations are the same as the $PM_{2.5}$ Targets Report (See *Appendix B: Supporting Information for Testing Protocols*).

Effect of Relative Humidity (RH) recommendations are the same as the $PM_{2.5}$ Targets Report (See *Appendix B: Supporting Information for Testing Protocols*). Although the RH test conditions are the extent of these protocols, we acknowledge that additional testing at higher RH levels may be valuable considering the environments where PM_{10} sensors are deployed. Testers may choose to conduct additional enhanced testing under additional RH conditions.

Effect of Temperature (T) recommendations are the same as the $PM_{2.5}$ Targets Report (See *Appendix B: Supporting Information for Testing Protocols*). Although the T test conditions are the extent of these protocols, we acknowledge that additional testing at higher T levels may be valuable considering the environments where PM_{10} sensors are deployed. Testers may choose to conduct additional enhanced testing under additional T conditions.

Detection Limit (DL) recommendations are the same as the $PM_{2.5}$ Targets Report (See *Appendix B: Supporting Information for Testing Protocols*).

Drift testing protocols were informed by other sensor evaluations. The low concentration drift test uses a low PM_{10} concentration test rather than zero air because some sensors do not provide true measurements for zero air. This concentration was chosen to be above the DL of most air sensors (as specified by sensor manufacturers). The mid concentration drift test setpoint slightly exceeds the AQI green/yellow (good/moderate) breakpoint where it would become important for health messaging. A 60-day test, or aging, period was a compromise between a short-term (24-hour) test period and a longer term (e.g., months, years) test period. 60-days is presumed important to measure potential changes in sensor performance over a longer timeframe but not too long as to be an undue burden and hinder completion of a sensor performance test. These protocols recommend that sensors be aged by 60 days of continual operation in outdoor, ambient air to be most representative of routine operation with good variation in T and RH conditions.

Accuracy at High Concentrations testing protocols were informed by other sensor evaluations. This testing protocol recommends additional calibration points at higher AQI breakpoints (in addition to low and mid concentration) but prescribes that this test be conducted last as some literature indicates that exposure to high PM concentrations can accelerate sensor aging and reduce sensor response, both of which can damage the sensor. The high PM_{10} concentration setpoints were determined based on levels that slightly exceed the AQI yellow/orange (moderate/unhealthy for sensitive groups) and orange/red (unhealthy for sensitive groups/unhealthy) breakpoints where sustained high concentration measurements would be important for health messaging. The high PM_{10} concentration setpoint of 200 $\mu g/m^3$ is slightly above the current primary (health-based), 24-

hour National Ambient Air Quality Standard (NAAQS) of 150 μg/m^3. While it is possible for ambient PM_{10} concentrations to exceed the higher concentration setpoint (300 μg/m^3), this setpoint was selected as some exposure chambers may have a challenge generating PM concentrations above this level and/or repeated exposure to high concentrations may require more extensive and frequent maintenance of the FEM.

Appendix C: Supporting Information for Performance Metrics

Base testing and enhanced testing performance metrics for PM_{10} air sensors are largely the same as those for $PM_{2.5}$ air sensors. Please refer to the $PM_{2.5}$ Targets Report for details on the metrics. New as well as supplemental information for PM_{10} sensors is provided below.

Base and Enhanced Testing Performance Metrics

Exploring Effect of Meteorology. New to this report is an exploration of how wind speed impacts sensor performance. Because air sensors are often low power and low flow, they may not efficiently aspirate coarse particles. The aspiration efficiency may further decrease if the flow through the sensor must also work against wind moving particles in another direction. The effects may be most pronounced when wind acts directly opposite the flow into the sensor so wind direction may have an observable influence. Current evidence suggests that the magnitude of this impact is dependent on sensor design, orientation to the wind, and wind speed. Understanding the impact of wind on sensor performance may better inform sensor deployment considerations (e.g., where it will be deployed, what orientation should be used based on prevailing wind directions) and under what wind conditions sensor data may be less reliable.

$PM_{2.5}/PM_{10}$ Ratio. This is a new performance metric included to understand the size distribution of particles in different conditions. Lower $PM_{2.5}/PM_{10}$ ratios indicate that larger particles are more dominate, whereas higher ratios indicate that smaller particles are more dominate. Conditions where $PM_{2.5}$ and $PM_{2.5}/PM_{10}$ ratios are more variable can inform whether PM_{10} sensors are measuring coarse PM.

Additional Enhanced Testing Performance Metrics

Accuracy at High Concentrations. This performance metric is included to evaluate the suitability of a PM_{10} sensor for specific NSIM applications where high PM_{10} concentrations are expected. These NSIM applications can include, for example, dust storms or windblown dust, construction or demolition activities, agriculture, or fires. This metric helps determine the degree to which sensor measurements can be trusted in high PM_{10} environments.

Appendix D: Supporting Information for Target Values

The PM_{10} target values were informed by the following:

- The FRM/FEM certification program (Table D-1);
- The U.S. EPA's findings on air sensor evaluations (Table D-2);
- AQ-SPEC air sensor field evaluations (Table D-3);
- Peer-reviewed literature reporting data quality levels (Table D-4); and
- Sensor standards/certification programs in development by other organizations (Table D-5).

Table D-1. Performance Requirements for PM_{10} FRM/FEM Regulatory Monitors

Performance Attribute	Specification for Regulatory Monitoring	Notes (Based on 40 CFR Part 53 Subpart C and 40 CFR Part 50)
Accuracy/Uncertainty	R^2: $\geq$ 0.97 Slope: 1 ± 0.10 Intercept: 0 ± 5 µg/m^3	
Measurement Range	15-300 µg/m^3	Referred to as 'acceptable concentration range (R_j)'.
Detection Limit	The lower limit of the mass concentration range is determined by the repeatability of filter tare weights, assuming the nominal air sample volume for the sampler.	Referred to as 'lower detection limit'.
Precision	CV_{conc}: ≤ 7% (for PM_{10} concentrations > 80 µg/m^3) SD: ≤ 5 µg/m^3 (for PM_{10} concentrations < 80 µg/m^3) Root mean square: 10%	CV_{conc} represents the concentration coefficient of variation.

Table D-2. Summary of U.S. EPA PM_{10} Sensor Evaluations

Averaging Interval	Sensor Manufacturer/ Model	Concentration Range ($\mu g/m^3$)		Slope	Intercept ($\mu g/m^3$)	R^2
		Sensor	Reference			
24-hours	Alphasense/OPC-N2	5-80	10-60	0.02	25.03	< 0.01
1-hour	Alphasense/OPC-N2	5-70	5-50	0.18	22.25	0.01

Note: For U.S. EPA evaluations, SD and RMSE values were not calculated. Slope and intercept were determined with reference data on the x-axis and sensor data on the y-axis as recommended in this report.

Table D-3. Summary of AQ-SPEC PM_{10} Sensor Field Evaluation Results (1-hour Average)

Sensor Manufacturer/ Model/Version	Concentration Range (µg/m³)	Slope*	Intercept* (µg/m³)	$R^{2,*}$	SD (µg/m³)	RMSE (µg/m³)
Aeroqual/AQY/v1.0	0-350	0.50 to 0.64	-17.08 to -8.02	0.40-0.74	-	-
Aeroqual S500-PM	0-250	1.24 to 1.52	-35.83 to -18.78	0.20-0.30	-	-
Airly	0-350	1.20 to 1.28	-45.76 to -24.54	0.18-0.40	21.9	27.6-32.5
AirThinx/IAQ	0-430	1.63 to 2.00	-68.95 to -46.35	0.05-0.07		33.46-39.54
Alphasense/OPC-N2	0-125	-	-	0.42-0.60	-	-
Alphasense/OPC-N3	5-375	0.80 to 1.23	-32.40 to -9.45	0.28-0.53	-	-
Alphasense/OPC-R2	0-215	0.95 to 1.16	-12.16 to -2.99	0.71-0.90	-	-
Applied Particle Technology/MINIMA	0-350	0.67 to 0.92	-33.99 to -23.09	0.23-0.42	13.5	37.8-46.8
AQMesh/v3.0	0-280	0.36 to 0.64	-9.02 to 6.05	0.29-0.70	-	-
AS-LUNG/Air Quality Station	5-180	1.56 to 1.82	-59.02 to 52.15	0.26-0.28	-	-
AS-LUNG/Portable	0-500	1.65 to 1.79	-72.59 to -67.67	0.09-0.10	-	-
Atmotube/Pro	0-275	0.85 to 1.13	-30.35 to -17.52	0.18-0.26	-	-
Blues Wireless/Airnote	5-120	1.05 to 2.45	-62.61 to -37	0.05-0.13	5.1	23.0-36.9
Redspira	0-350	1.06 to 1.46	-51.91 to -33.44	0.20-0.40	19.8	38.1-41.7
Davis Instruments/Airlink	0-150	1.48 to 2.16	-49.05 to -40.05	0.26-0.33	13.8	15.5-29.7
Dylos/DC1700-PM	5-490	4.41 to 7.48	-317.19 to -129.14	0.06-0.21	-	-
Ecomesure/EcomSmart	0-150	2.61 to 4.80	-103.07 to -62.47	0.10-0.30	26.7	21.7-28.8
Elitech/Temtop LKC-1000S+	0-180	1.74 to 1.90	-29.10 to -25.98	0.35-0.37	-	-
Elitech/Temtop M2000	0-250	0.75 to 1.01	-16.11 to -8.49	0.24-0.33	-	-
Elitech/Temtop PMD 351	5-120	1.23 to 2.45	-41.17 to -26.26	0.28-0.48	-	-
Fablab/Smart Citizen Kit/v2.1	0-625	1.83 to 3.14	-182.25 to -102.19	0.04-0.10	-	-
HabitatMap/AirBeam2	10-300	2.63 to 8.03	-345.33 to -102.60	0.006-0.01	-	-
HabitatMap/AirBeam3	0-175	0.62 to 1.04	-21.24 to -14.72	0.21-0.28	7.8	26.6-31.5
Igienair/Zaack AQI	0-350	1.44 to 2.11	-30.84 to -30.76	0.70-0.86	46.0	20.1-38.7
IQAir/AirVisual Pro	0-275	1.10 to 1.12	-26.55 to -18.21	0.25-0.41	-	-
IQAir/AirVisual Outdoor	0-215	0.66 to 0.93	5.40 to 16.45	0.41-0.66	11.8	13.3-17.9
Kaiterra/Laser Egg 2+	0-150	1.11 to 1.66	-31.13 to -14.53	0.12-0.31	-	-
Kunak/Air A10	0-245	0.48 to 0.61	-4.69 to -1.95	0.57-0.76	1.73	21.57-27.07
Liveable Cities/SLX-$PM_{2.5}$	0-275	0.80 to 1.17	-9.56 to -3.66	0.80-0.89	5.82	11.7-23.2
Lunar Outpost/Canary-S	5-280	1.61 to 2.12	-89.45 to -56.94	0.09-0.18	0.82	29.26-35.77
Magnasci SRL/uRADMonitor A3/HW105	0-180	0.72 to 1.18	-32.31 to -17.12	0.15-0.41	-	-

Sensor Manufacturer/ Model/Version	Concentration Range (µg/m^3)	Slope*	Intercept* (µg/m^3)	$R^{2,*}$	SD (µg/m^3)	RMSE (µg/m^3)
Magnasci SRL/uRADMonitor INDUSTRIAL/HW103	0-195	0.78 to 1.42	-38.23 to -16.42	0.12-0.35	-	-
MagnasciSRL/uRADMonitor SMOGGIE-PM/v1.101	0-250	0.97 to 2.32	-70.47 to -18.45	0.04-0.10	-	-
MetOne (ES-405)	0-175	0.95 to 1.18	-6.37 to 0.64	0.71-0.96	20.6	5.6-12.3
Oizom/Polludrone Smart	5-200	2.04 to 2.16	-55.71 to -50.40	0.34-0.35	21.5	17.3-19.1
Origins/Laser Egg	0-230	2.39 to 3.01	-120.73 to -93.44	0.03 to 0.05	-	-
Piera Systems/Canaree R1	0-170	1.65 to 2.38	-49.63 to -39.55	0.30-0.37	20.0	21.4-26.8
Plume Labs/Flow 2	0-250	2.78 to 11.25	-330.29 to -67.73	0.003-0.09	-	-
PM Monitor/iMonPM	0-150	0.90 to 1.17	-15.34 to -11.96	0.39-0.66	13.4	16.5-20.9
Purple Air/PA-I	0-100	1.46 to 1.61	-37.58 to -19.37	0.32-0.45	-	-
PurpleAir/PA-I-Indoor	0-100	0.03 to 0.07	-0.69 to -0.39	0.36-0.46	-	-
PurpleAir/PA-II	0-100	1.34 to 1.87	-20.34 to -10.71	0.61-0.72	-	-
PurpleAir/PA-II-FLEX	0-150	1.09 to 1.61	-32.16 to -26.97	0.22-0.42	14.4	21.0-27.0
Qingping Air Monitor Lite	0-125	0.60-0.65	-8.82 to -5.95	0.40-0.45	7.5	22.1-24.7
QuantAQ/MODULAIR-PM	0-375	0.60 to 0.85	-9.30 to -1.18	0.47-0.79	14.8	19.8-28.4
SailBri Cooper/SCI-901	0-150	0.83 to 1.53	-5.95 to 9.41	0.65-0.86	16.3	6.4-17.2
Sensirion/SPS30	0-150	0.78 to 1.68	-29.87 to -17.60	0.11-0.25	-	-
Strop de aer	10-150	0.65 to 1.30	-29.41 to -16.56	0.15-0.29	6.0	22.4-33.5
Tera Sensor/NextPM	0-375	0.63 to 0.99	-25.12 to -12.30	0.35-0.67	17.4	34.6-37.8
TSI/BlueSky		0.44 to 0.68	-16.48 to -6.80	0.14 to 0.21	-	-
Wicked Device/Air Quality Egg/2018 Model	5-125	1.75 to 1.90	-61.01 to -58.83	0.11-0.12	-	-
Wicked Device/Air Quality Egg/2022 Model	0-275	1.05 to 1.61	-38.26 to -20.20	0.30-0.54	26.1	27.9-30.2
Wicked Device/Air Quality Egg/Version 2	0-130	0.72 to 1.20	-26.27 to -12.59	0.07-0.40	-	-

Note: These field evaluations were conducted at the Rubidoux air monitoring station in Riverside, CA. Evaluations are current as of 03/30/2023.
*AQ-SPEC presents graphical results with reference instrument measurements on the y-axis and sensor measurements on the x-axis, which is the reverse of the recommended method in this report. The results shown in this Table mathematically manipulate the equations reported by AQ-SPEC to present slopes and intercepts in a similar form to that recommended in this report. It should be noted that these results are approximate as performing a least squares regression on the data with the x-axis and y-axis variables switched will produce different results. As a result, sensors highlighted in gray ($R^2 \geq 0.75$) were used to inform the bias (slope and intercept) targets. All results were used to inform the targets for R^2, SD, and RMSE.

Table D-4. Summary of AQ-SPEC PM_{10} Sensor Field Evaluation Results (24-hour Average)

Sensor Manufacturer/ Model/Version	Concentration Range (µg/m^3)	Slope*	Intercept* (µg/m^3)	$R^{2,*}$	SD (µg/m^3)	RMSE (µg/m^3)
Aeroqual/AQY/v1.0	15-100	0.49 to 0.64	-16.68 to -7.83	0.59-0.83	-	-
Aeroqual S500-PM	10-65	1.43 to 1.84	-47.63 to -25.50	0.21-0.45	-	-
Airly	5-100	1.17 to 1.69	-46.30 to -23.44	0.27-0.47	15.4	20.50-21.4
AirThinx/IAQ	5-130	1.56 to 2.30	-65.35 to -54.04	0.05-0.10	-	-
Alphasense/OPC-N2	-	-	-	-	-	-
Alphasense/OPC-N3	10-85	0.68 to 1.85	-54.21 to -7.73	0.23-0.74	-	-
Alphasense/OPC-R2	5-150	0.73 to 1.13	-3.02 to 2.75	0.83-0.92	-	-
Applied Particle Technology/MINIMA	5-100	0.63 to 0.85	-30.36 to -21.04	0.40-0.53	9.2	37.6-42.5
AQMesh/v3.0	5-60	0.42 to 0.98	-19.24 to 2.99	0.21-0.81	-	-
AS-LUNG/Air Quality Station	-	-	-	-	-	-
AS-LUNG/Portable	-	-	-	-	-	-
Atmotube/Pro	5-75	0.83 to 1.16	-31.39 to -17.00	0.31-0.39	-	-
Blues Wireless/Airnote	15-55	1.76 to 54.64	-1570.05 to -66.54	0-0.08	3.7	21.9-35.1
Redspira				0.35-0.52	13.6	31.2-35.3
Davis Instruments/Airlink	5-65	1.47 to 2.42	-56.10 to -45.25	0.23-0.33	10.0	12.9-28.0
Dylos/DC1700-PM	5-110	3.47 to 217.39	-10499.35 to -82.32	0-0.39	-	-
Ecomesure/EcomSmart	10-60	2.79 to 4.81	-106.57 to -70.82	0.10-0.37	21.7	15.4-21.9
Elitech/Temtop LKC-1000S+	0-70	1.63 to 1.76	-25.47 to -23.13	0.49-0.51	-	-
Elitech/Temtop M2000	0-55	0.63 to 0.85	-10.50 to -5.00	0.42-0.53	-	-
Elitech/Temtop PMD 351	5-60	1.34 to 3.14	-60.73 to -30.37	0.21-0.51	-	-
Fablab/Smart Citizen Kit/v2.1	20-150	1.90 to 3.06	-198.61 to -106.64	0.10-0.23	-	-
HabitatMap/AirBeam2	25-85	-28.09 to 8.20	-430.92 to 1298.01	0.00-0.09	-	-
HabitatMap/AirBeam3	5-60	0.64 to 1.19	-25.68 to 15.35	0.23-0.37	6.1	23.0-28.8
Igienair/Zaack AQI	5-100	1.51 to 1.90	-20.61 to –12.98	0.66-0.85	26.8	15.4-29.1
IQAir/AirVisual Pro	15-75	1.58 to 1.64	-50.49 to –38.95	0.16-0.25	-	-
IQAir/AirVisual Outdoor	5-75	0.64 to 0.88	6.99 to 17.05	0.51-0.76	8.4	8.3-12.1
Kaiterra/Laser Egg 2+	5-55	1.18 to 1.89	-44.78 to –15.18	0.11-0.33	-	-
Kunak/Air A10	10-95	0.43 to 0.60	-4.55 to -1.02	0.80-0.86		
Liveable Cities/SLX-$PM_{2.5}$	2-150	0.69 to 1.15	-7.45 to –0.88	0.88-0.95	-	-
Lunar Outpost/Canary-S	30-100	2.64 to 3.38	-140.92 to –117.50	0.05-0.09	-	-
Magnasci SRL/uRADMonitor A3/HW105	0-110	0.54 to 0.92	-22.73 to –10.41	0.23-0.53	-	-

Sensor Manufacturer/ Model/Version	Concentration Range (µg/m^3)	Slope*	Intercept* (µg/m^3)	$R^{2,*}$	SD (µg/m^3)	RMSE (µg/m^3)
Magnasci SRL/uRADMonitor INDUSTRIAL/HW103	0-110	0.58 to 1.23	-31.91 to –9.21	0.15-0.50	-	-
MagnasciSRL/uRADMonitor SMOGGIE-PM/v1.101	5-65	0.81 to 2.43	-75.04 to –12.60	0.04-0.24	-	-
MetOne (ES-405)	5-75	0.93 to 1.31	-8.44 to –1.57	0.75-0.98	14.9	4.4-8.9
Oizom/Polludrone Smart	25-82	2.56 to 2.85	-86.80 to –74.88	0.27-0.34	15.8	13.1-14.2
Origins/Laser Egg	15-65	3.96 to 6.75	-293.85 to –166.67	0.01-0.03	-	-
Piera Systems/Canaree R1	5-60	1.88 to 3.12	-67.48 to –47.79	0.22-0.42	16.7	17.7-23.9
Plume Labs/Flow 2	9-55	1.99 to 7.23	-221.50 to –42.36	0.00-0.40	-	-
PM Monitor/iMonPM	10-60	0.90 to 1.51	-25.91 to –13.26	0.30-0.74	9.7	13.5-19.0
Purple Air/PA-I	5-65	1.39 to 1.66	-37.57 to –20.60	0.46-0.50	-	-
PurpleAir/PA-I-Indoor	-	-	-	-	-	-
PurpleAir/PA-II	0-70	1.30 to 1.79	-20.17 to –8.56	0.68-0.78	-	-
PurpleAir/PA-II-FLEX	10-60	1.13 to 2.04	-44.14 to –29.70	0.20-0.50	11.5	17.6-24.6
Qingping Air Monitor Lite	5-60	0.54 to 0.65	-8.80 to –4.24	0.42-0.45	5.2	19.0-22.6
QuantAQ/MODULAIR-PM	8-95	0.69 to 1.06	-20.60 to –2.32	0.37-0.81	9.5	16.3-24.3
SailBri Cooper/SCI-901	10-60	0.80 to 1.71	-10.13 to 7.39	0.67-0.95	12.0	2.7-15.7
Sensirion/SPS30	5-65	0.63 to 1.61	-20.97 to –12.64	0.10-0.30	-	-
Strop de aer	20-80	0.70 to 1.49	-35.71 to –17.38	0.20-0.53	4.4	20.0-31.0
Tera Sensor/NextPM	15-100	0.74 to 1.52	-50.24 to –18.10	0.32-0.72	21.4	29.5-33.9
TSI/BlueSky	0-65	0.33 to 0.55	-11.58 to –3.91	0.28-0.35	-	-
Wicked Device/Air Quality Egg/2018 Model	-	-	-	-	-	-
Wicked Device/Air Quality Egg/2022 Model	2-145	1.01 to 1.74	-42.14 to -17.90	0.27-0.62	22.5	23.1-24.9
Wicked Device/Air Quality Egg/Version 2	5-70	0.68 to 1.37	-30.49 to -12.62	0.07-0.63	-	-

Note: These field evaluations were conducted at the Rubidoux air monitoring station in Riverside, CA. Evaluations are current as of 03/30/2023.
*AQ-SPEC presents graphical results with reference instrument measurements on the y-axis and sensor measurements on the x-axis, which is the reverse of the recommended method in this report. The results shown in this Table mathematically manipulate the equations reported by AQ-SPEC to present slopes and intercepts in a similar form to that recommended in this report. It should be noted that these results are approximate as performing a least squares regression on the data with the x-axis and y-axis variables switched will produce different results. As a result, sensors with $R^2 \geq 0.75$ were used to inform the bias (slope and intercept) targets. All results were used to inform the targets for R^2, SD, and RMSE.

Table D-5. Summary of Literature Reviews of PM_{10} Sensor Evaluations (1-hour average)

Source (Location)	Sensor Manufacturer / Model	Concentration Range[§] (µg/m³)	$PM_{2.5}/PM_{10}$ Ratio Range	CV (%)	Slope[*]	Intercept[*] (µg/m³)	$R^{2,*}$	RMSE (µg/m³)
Crilley et al., 2018 (Birmingham Elms Road Observatory Station, UK)	Alphasense/ OPC-N2	0-250			2.06[†]		0.67[†]	
Crilley et al., 2018 (Birmingham Automatic and Rural Urban Network, UK)	Alphasense/ OPC-N2	0-55		22	3.3 (2.5 to 4.4)[†]		0.67 (0.64 to 0.69)[†]	
Feinberg et al., 2018 (Denver, CO, USA)[**]	Alphasense/ OPC-N2	19.6[#]			0.37 (0.12 to 0.54)	1.59 (-1.06 to 2.98)	0.24 (0.04 to 0.46)	
Mukherjee et al., 2017 (Cuyama Valley, CA, USA)	Alphasense/ OPC-N2	0-250			0.19 (0.13 to 0.23)	2.38 (0.32 to 4.62)	0.68 (0.53 to 0.81)	
Castell et al., 2017 (Oslo Kirkeveie, Norway)	Environmental Instruments/ AQMesh				1.70 (1.30 to 2.10)		0.31 (0.04 to 0.53)	18.5 (13.8 to 64.4)
Castell et al., 2017 (Oslo Åkebergveien, Norway)	Environmental Instruments/ AQMesh				2.10	-1.30	0.68	
Kaur and Kelly, 2022 (Salt Lake Valley, UT, USA)	Alphasense/ OPC-N3	1.9 to 274	0.056 to 1		0.92	-0.42	0.87	12.4
	Plantower/ PMS 5003	1.9 to 274	0.056 to 1		0.06 (0.03 to 0.09)	2.66 (1.97 to 3.38)	0.29 (0.13 to 0.48)	35.2 (33.7 to 36.3)
Kaur and Kelly, 2022 (Environmental Quality Station, Salt Lake Valley, UT, USA)	Plantower/ PMS 5003	0.9 to 327	0.026 to 1		0.05 (0.03 to 0.07)	3.70 (3.03 to 4.35)	0.21 (0.096 to 0.34)	43.8 (42.6 to 45.1)
Kaur and Kelly, 2022 (Residential site, Salt Lake Valley, UT, USA)	Alphasense/ OPC-N3	0 to 225			1.39	-1.08	0.94	17.7
	Plantower/ PMS 5003	0 to 175			0.08 (0.06 to 0.10)	3.15 (2.70 to 3.25)	0.42 (0.33 to 0.48)	34.15 (33.5 to 34.6)

[§]As measured by the reference instrument; [*]Values represent mean with range in parenthesis, where available; [†]Calculated using PM_{10} median concentrations; [**]Reported on 24-hour average correlation results only with an average R^2 of 0.35 and range of 0.03 to 0.55; [#]Average concentration

Table D-6. Summary of Resources Used to Inform PM_{10} Target Values

		Precision (SD)	Slope	Intercept ($\mu g/m^3$)	R^2	RMSE ($\mu g/m^3$)
Evaluation Data*	**Range**					
	24-hour	4.40 to 26.80	0.50 to 1.50	-13.90 to 14.68	0.50 to 0.98	2.70 to 42.50
	1-hour	0.82 to 26.10	0.56 to 1.44	-12.16 to 9.41	0.76 to 0.98	5.60 to 38.70
	Average					
	24-hour	13.38	0.96	-2.55	0.86	21.30
	1-hour	13.73	1.01	-3.73	0.74	20.97
	Median					
	24-hour	11.50	0.96	-2.53	0.86	21.63
	1-hour	14.40	0.97	-4.09	0.74	19.95
People's Republic of China MEE		Not discussed/ not listed	Not discussed/ not listed	Not discussed/ not listed	$\geq$ 0.64	Not discussed/ not listed
This Report	Target (Base Testing Only)	≤ 5 $\mu g/m^3$ (SD) OR ≤ 30% (CV)	1.0 ± 0.35	-10 ≤ b ≤ 10	≥ 0.70	RMSE ≤ 14 $\mu g/m^3$ or NRMSE ≤ 30%†

*Data with slopes outside of 0.5 to 1.5 were removed. The intercept was not considered if the slope was discarded. Data with R^2 less than 0.5 was not considered. AQ-SPEC data, where the slope and intercept data were calculated by mathematical manipulation, was only considered if $R^2 \geq 0.75$.
†A sensor will meet this target if either the RMSE or NRMSE meet this criterion. See Appendix D for further discussion.

Rationale is provided below for the recommended target values for each of the recommended performance metrics.

Precision: ≤ 5 $\mu g/m^3$ (SD) OR ≤ 30% (CV) [Same as $PM_{2.5}$ sensor target value]

Strong precision between sensors is defined by sensors behaving similarly to environmental conditions therefore minimizing variation between individual sensor devices and measurement noise. The precision target is the same target recommended in the $PM_{2.5}$ Targets Report. Although the SD target for precision is similar to the performance requirement for PM_{10} FRM/FEM regulatory monitors (for PM_{10} concentrations < 80 $\mu g/m^3$), strong precision is essential for a wide range of NSIM applications, especially popular applications where concentration measurements from one sensor must be compared to that of another. The strongest precision would yield a SD of 0 $\mu g/m^3$ or a CV of 0% meaning that all sensors respond identically to environmental conditions. It is important to note that precision is just one performance metric and needs to be considered in conjunction with other metrics to better understand sensor performance. Sensor evaluations programs and PM_{10} sensor evaluations both note that there may be slight differences between measurements from different devices, which is accommodated for within a range of precision target values.

Slope (Bias): 1.0 ± 0.35 [Same as $PM_{2.5}$ sensor target value]

The target value for the slope component of bias is 1.0 ± 0.35. This target is the same target recommended in the $PM_{2.5}$ Targets Report. The slope target indicates that PM_{10} sensors show roughly the same increase or decrease in PM_{10} concentrations as the reference instrument, making changes in concentration comparable between the two different devices. This is critical for NSIM applications where relative difference, or the amount of change, is important. The proposed target value prescribes a confidence interval to assist testers in evaluating sensor performance.

Intercept (Bias): -10 ≤ b ≤ 10 μg/m^3

The target value for the intercept (b) component of bias is 0 ± 10 μg/m^3. This target is twice the target recommended in the $PM_{2.5}$ Targets Report. Typically, PM_{10} concentrations reported by regulatory monitoring sites range between equal to and 4 times that of $PM_{2.5}$ concentrations suggesting a larger intercept may be appropriate. Review of recent AQ-SPEC evaluations suggests that a wider intercept target that is approximately 2 times that of the $PM_{2.5}$ target is more appropriate and more readily achievable by current technology. This target ensures that low concentration measurements are still meaningful, and that system error is minimized. The proposed target value prescribes a confidence interval to assist testers in evaluating sensor performance. It should be noted that while the intercept target for bias is similar to the intercept performance requirement for PM_{10} FRM/FEM regulatory monitors, the slope is considered a more important performance metric. If the slope improves, the intercept tends to improve.

R^2 (Linearity): ≥ 0.70 [Same as $PM_{2.5}$ sensor target value]

The target value for linearity is ≥ 0.70. This target is the same target recommended in the $PM_{2.5}$ Targets Report. Higher R^2 values are associated with closer agreement and better linearity between two data sets being compared. R^2 should be considered in conjunction with other performance metrics because high linearity does not necessarily indicate perfect agreement between data sets (e.g., two datasets can have an R^2 close to 1 with a linear regression slope of 2, as a result of different absolute concentration values between the data sets). Care should be taken when interpreting R^2, as poor linearity can be caused by a number of reasons such as a non-linear relationship, outliers, or lack of precision in the sensor or reference instrument. Linearity can also be strongly influenced by just a few high concentration measurements.

RMSE (Error): ≤ 14 μg/m^3 (RMSE) or ≤ 30% (NRMSE) [NRMSE same as $PM_{2.5}$ target value]

The target value for RMSE is ≤ 14 μg/m^3. This target is twice as large as the RMSE target recommended in the $PM_{2.5}$ Targets Report. While there is limited reported data on which to base this target value, the RMSE target was selected to reflect ambient PM_{10} concentrations which are often higher than $PM_{2.5}$ concentrations. RMSE quantifies the random disagreement between the pollutant concentrations reported by the sensor and the reference instrument, thus values closer to zero indicate better agreement and less uncertainty in the measurement. This is an important metric for NSIM applications where sensor and reference instrument concentrations need to be compared. RMSE is sensitive to data points with large differences between sensor and reference instrument

concentrations. Care should be taken to use the definition and recommended calculation for RMSE that is provided in the $PM_{2.5}$ Targets Report (*see Sections 3.1.5 and 3.2.5*). A target value for NRMSE of ≤ 30% is also included to account for testing conditions where the ambient PM_{10} concentrations may be much higher than typical U.S. ambient levels (e.g., wildfires, dust storms). Under these conditions, the RMSE metric may exceed the target because concentrations are high, but the NRMSE may meet this target instead. Both RMSE and NRMSE may be reported, but the tester may report the more favorable result (the result meeting or closest to this target metric). The sensor will meet this target if either the RMSE or NRMSE meet this criterion.

Effect of Relative Humidity (RH): No Target Value Established

The effect of RH on sensor measurements is an important performance metric for all NSIM applications especially because RH varies across the U.S. and can change rapidly throughout the day. Many studies have shown that currently available PM_{10} sensors are affected by RH (e.g., Jayaratne et al., 2018; Rogulski and Badyda, 2020; Tagle et al., 2020; AQ-SPEC PM sensor evaluations). Because PM_{10} sensors do not typically heat the inlet sample stream to drive off water vapor, high moisture content in the ambient, outdoor air can change the refractive indices of particulate matter (PM) in the sample stream and can also lead to hygroscopic growth of particles (e.g., at high RH levels, mist or fog could be detected as PM). Literature sources attempting to quantify the effects of RH did not report results in a consistent manner thus, a target level has not been established. The protocols outlined in this report request that testers quantify the influence of RH in a systematic way. This work may inform the future establishment of a target value.

Effect of Temperature (T): No Target Value Established

The effect of ambient T on sensor measurements is an important performance metric for all NSIM application areas because T varies significantly throughout the day, across different seasons, and across the U.S. Considering the very limited performance data available, this report does not establish a target value for this metric. The protocols outlined in this report request that testers quantify the influence of T in a systematic way which may inform the future establishment of a target value.

Drift: No Target Value Established

While little to no drift is ideal, the available information and literature suggests that measurements from many air sensors may drift over time. The literature suggests that drift may occur abruptly or steadily as the sensor ages, on the order of days, months, or years. The rate of drift is currently understood to be highly variable, may depend on the concentrations experienced, and may still occur even if the sensor is not being used. The rate and degree of drift has not been systemically quantified in the literature. Currently, there has been little testing on drift in air sensors on the 60-day scale therefore, this report does not establish a target value for drift. The protocols outlined in this report request that testers quantify the influence of drift in a systematic way after 60-days of operation in an outdoor, ambient environment. A 60-day evaluation period is recommended to reduce the burden on testers. These results will help establish whether drift can be observed within a 60-day period and may inform the future establishment of a target value.

Accuracy at High Concentrations: No Target Value Established

Many sensor manufacturers/developers claim the ability to accurately measure PM_{10} concentrations at high concentrations. Discussions with groups that evaluate air sensors suggests that sensor measurements are more likely to differ from reference instrument measurements at high concentrations. Few field measurements are made at high concentrations because they occur less frequently. Understanding how accurately a sensor performs during higher PM_{10} concentrations is important for areas that experience such conditions, for NSIM applications focused on exceptional events (e.g., wildfires, dust storms), and for verifying whether potential corrections still apply at higher concentrations. A target value has not been established at this time.

Appendix E: Checklist for Base Testing

Data Collected Before Base Testing (Sections 2.1.1 through 2.1.3)

- ☐ Testing Organization(s) Name and Contact Information [email, phone number, and/or website]
- ☐ Testing location [City, State; Latitude, Longitude; AQS site ID (*if applicable*)]
- ☐ Information about air sensor spacing relative to the FRM/FEM monitor and other air sensors
- ☐ Information about any weather-protective shelter/enclosure used (*if applicable*)
- ☐ Relative humidity (RH), temperature (T), and FRM/FEM monitor information, including:

Item (*as applicable*)	**RH Monitor**	**T Monitor**	**PM_{10} FRM/FEM Monitor**	**$PM_{2.5}$ FRM/FEM Monitor**	**Other Reference Monitor (s)**
Manufacturer/Model					
Firmware Version					
Parameter(s) Measured and Units					
Sampling Time Interval					
Manufacturer Specification Sheet					
Copy of Calibration Certificate					

- ☐ Air sensor equipment information, including:

Item (as applicable)		**Sensor 1**	**Sensor 2**	**Sensor 3**
General Information	Manufacturer/Model			
	Firmware Version			
	Serial/Identification Number			
	Parameter(s) Measured and Units			
	Sampling Time Interval			
	Manufacturer Specification Sheet			
Data Storage Information	Where the data are stored			
	Where the data are transmitted			
	Form of data stored			
Data Correction Approach	Procedure used to correct data			
	If data correction does not change or is static, record approach			
	If data correction does change or is dynamic, record approach			
Data Analysis/Correction Script	Script used and version			
Final Data Reported	Location of final data			
	Format of final data			

- ☐ Photo(s) of entire equipment set up at test site

Data Collected During Base Testing (Section 2.1.4)

- ☐ Deployment number and sampling timeframe
- ☐ Dates for calibration and one-point flow rate verification check on the FRM/FEM monitor
- ☐ At least 30 consecutive days of measurements
- ☐ Description of QC criteria (*as applicable*)
- ☐ Time, dates, description, and rationale for any of the following (*as applicable*): 1) maintenance, 2) missing or invalidated data, and 3) any other issue(s) impacting data collection

Appendix F: Example Reporting Template for Base Testing

Testing Report – PM_{10} Base Testing
Manufacturer & Air Sensor Name

Deployment Number
Testing Organization
Contact Email / Phone Number
Date

Image of device during deployment

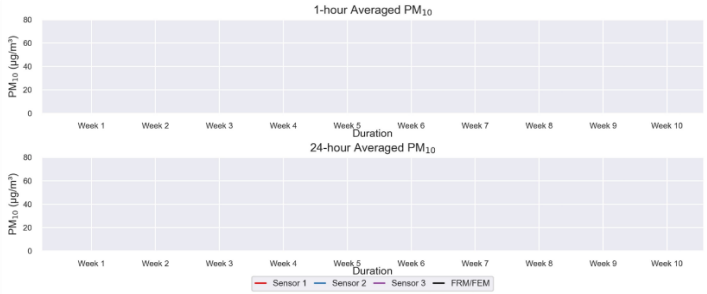
Deployment Details

Testing Organization and Site Information	
Testing organization (Name, Organization, Contact website / email / phone number)	
Testing location (City, State, Latitude and Longitude)	
AQS site ID	
Sampling timeframe (MM-DD-YY)	
Sensor data source	
Reference data source	

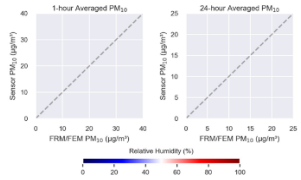
Sensor Information			
Manufacturer, model			
Device firmware version			
Sampling time interval			
Sensor serial numbers	Sensor 1	Sensor 2	Sensor 3
Issues encountered during deployment?	☐	Brief Summary of Issues	

FRM/FEM Information	
Manufacturer, model, designation	
Sampling time interval	
Date of calibration	
Date of flowrate verification check	
Description, date(s) of maintenance activities	

Time Series Plots

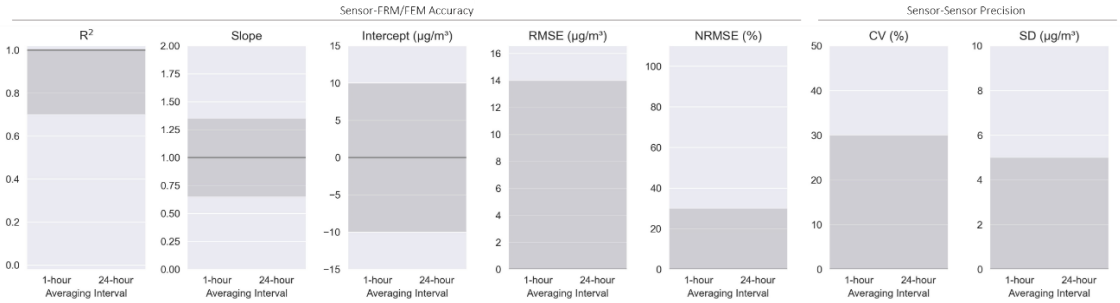


Scatter Plots: Comparison to FRM/FEM

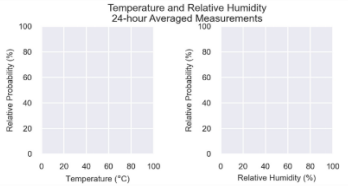


Range of FRM/FEM monitor concentrations over duration of test (µg/m³)	
Number of 24-hour periods in FRM/FEM monitor measurements with a goal concentration of ≥ 40 µg/m³	
Number of 1-hour periods in FRM/FEM monitor measurements with goal $PM_{2.5}/PM_{10}$ ratio of ≤ 0.4	

Performance Metrics*

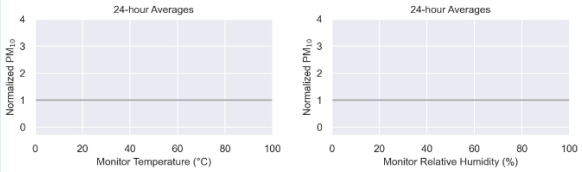


Meteorological Conditions During Deployment



Number of 24-hour periods outside sensor manufacturer-listed temperature operational range	
Number of 24-hour periods outside sensor manufacturer-listed relative humidity operational range	

Meteorological Influence



Number of paired, normalized concentration and temperature values	
Mean number of paired, normalized concentration and relative humidity values	

1:1
Sensor 1
Sensor 2
Sensor 3

*For evaluations with greater than three sensors, grouping individual sensor metrics into boxplots is recommended for displaying results. Note that this recommendation does not apply to metrics computed as a single value for all sensors over the whole evaluation group, such as RMSE, NRMSE, CV, and SD.

1

Testing Report – PM_{10} Base Testing
Manufacturer & Air Sensor Name

Deployment Number
Testing Organization
Contact Email / Phone Number
Date

Image of device during deployment

Tabular Statistics

Sensor-FRM/FEM Correlation

	Bias and Linearity						Data Quality			
	R^2		Slope		Intercept ($\mu g/m^3$)		Uptime (%)		Number of paired sensor and FRM/FEM concentration values	
	1-Hour ○○○	24-Hour ○○○	1-Hour ○○○	24-Hour ○○○	1-Hour ○○○	24-Hour ○○○	1-Hour ○○○	24-Hour ○○○	1-Hour	24-Hour
Metric Target Range	≥ 0.70	≥ 0.70	1.0 ± 0.35	1.0 ± 0.35	-10 ≤ b ≤ 10	-10 ≤ b ≤ 10	75%*	75%*	-	-
Sensor1										
Sensor2										
Sensor3										
Mean										

	Error			
	RMSE ($\mu g/m^3$)		NRMSE (%)	
	1-Hour ☆	24-Hour ☆	1-Hour ☆	24-Hour ☆
Metric Target Range	≤ 14.0	≤ 14.0	≤ 30.0	≤ 30.0
Deployment Value				

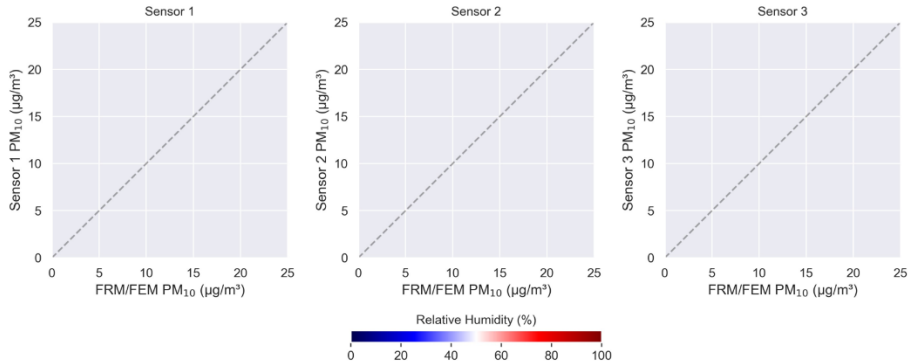
Device-specific metrics (computed for each sensor in evaluation)
○○○ Metric value for none of devices tested falls within the target range
●○○ Metric value for one of devices tested falls within the target range
●●○ Metric value for two of devices tested falls within the target range
●●● Metric value for three of devices tested falls within the target range

Single-valued metrics (computed via entire evaluation dataset)
☆ Indicates that the metric value is not within the target range
★ Indicates that the metric value is within the target range

Sensor-Sensor Precision

	Precision (between collocated sensors)				Data Quality	
	CV (%)		SD ($\mu g/m^3$)		Number of concurrent sensor concentration pairs	
	1-Hour ☆	24-Hour ☆	1-Hour ☆	24-Hour ☆	1-Hour	24-Hour
Metric Target Range	≤ 30.0	≤ 30.0	≤ 5.0	≤ 5.0	-	-
Deployment Value						

Sensor-FRM/FEM Scatter Plots



*This value is only a recommendation for ensuring data quality and is not included in the list of target values discussed in Section 4 of the Performance Testing Protocols, Metrics, and Target Values for Fine Particulate Matter Air Sensors document.

Testing Report – PM_{10} Base Testing
Manufacturer & Air Sensor Name

Deployment Number
Testing Organization
Contact Email / Phone Number
Date

Image of device during deployment

Supplemental Information

Additional documentation may be attached or linked to digital versions alongside this report. Such documentation may include field reports and observations during the testing period, maintenance logs for sensors and FRM/FEM monitors, standard operating procedures, and other documentation relevant to this testing report (see below for examples).

Supplemental Documentation	Attached	Description & URL or file path to documentation
Field observations	☐	
Maintenance logs	☐	
Standard operating procedure(s)	☐	
Photos of equipment setup and testing	☐	
Product specifications sheet(s)	☐	
Product manual(s)	☐	
Deployment issues	☐	
Data storage and transmission method	☐	
Data correction approach	☐	
Data analysis/correction scripts and version	☐	
Air Monitoring Station QAPP	☐	
Summary of FRM/FEM monitor QC checks	☐	
Other documents	☐	

3

Note: A fillable reporting template for base testing is also available with this report. See accompanying PowerPoint File.

Appendix G: Checklist for Enhanced Testing

Data Collected Before Enhanced Testing (Sections 2.2.1 and 2.2.2)

- ☐ Testing organization(s) name and contact information [email, phone number, and/or website]
- ☐ Testing address/location [City, State]
- ☐ Description of all chamber specifications and characterization
- ☐ Relative humidity (RH), temperature (T), and FEM monitor information, including:

Item (*as applicable*)	**RH Monitor**	**T Monitor**	**PM_{10} FEM Monitor**	**$PM_{2.5}$ FEM Monitor**	**Other Reference Monitor(s)**
Manufacturer/Model					
Firmware Version					
Parameter(s) Measured and Units					
Sampling Time Interval					
Manufacturer Specification Sheet					
Copy of Calibration Certificate					
Date of Calibration at Test Location					
Date of one-point flow rate verification check					

- ☐ Air sensor equipment information, including:

Item (as applicable)		**Sensor 1**	**Sensor 2**	**Sensor 3**
General Information	Manufacturer/Model			
	Firmware Version			
	Serial/Identification Number			
	Parameter(s) Measured and Units			
	Sampling Time Interval			
	Manufacturer Specification Sheet			
Data Storage Information	Where the data are stored			
	Where the data are transmitted			
	Form of data stored			
Data Correction Approach	Procedure used to correct data			
	If data correction does not change or is static, record approach			
	If data correction does change or is dynamic, record approach			
Data Analysis/Correction Scripts	Script used and version			
Final Data Reported	Location of final data			
	Format of final data			

- ☐ Photo(s) of entire equipment set up in exposure chamber (optional)

Data Collected During Enhanced Testing (Sections 2.2.3 through 2.2.7)

- ☐ All time-matched data points for each testing condition
- ☐ Description of QC criteria (*as applicable*)
- ☐ Time, dates, description, and rationale for any of the following (*as applicable*): 1) maintenance, 2) missing or invalidated data, and 3) any other issue(s) impacting data collection

Appendix H: Example Reporting Template for Enhanced Testing

Testing Report – PM_{10} Enhanced Testing
Manufacturer & Air Sensor Name

Testing Organization
Contact Email / Phone Number
Date

Image of device during chamber evaluation

Testing Details

Testing Organization and Contact Information	
Testing organization (Name, Organization Type)	
Contact Information (Website, Phone Number, Email)	

PM_{10} FEM Monitor Information	
Manufacturer, model	
Sampling time interval	
Date of calibration	
Date of flowrate verification check	

Sensor Information			
Manufacturer, model			
Device firmware version			
Sampling time interval			
Sensor serial numbers	#1	#2	#3
Manufacturer listed detection limit			
Manufacturer listed longevity, lifespan			
Manufacturer listed drift			

Attached Documentation	
FEM monitor documentation	
Description, date(s) of maintenance activities	☐
Sensor documentation	
Product specification sheet	☐
Product manual	☐
Description of parameters measured and units, and data flow	☐
Data storage and transmission method	☐
Data analysis/correction script and version	☐
Testing chamber documentation	
Description of chamber, particle dispenser system, and particle sizer instrument	☐

Effect of Relative Humidity (RH)

RH Monitor	Manufacturer	
	Model	

			Average RH (%)	Average T (°C)	Average FEM monitor PM_{10} concentration of test aerosol (µg/m³)	Average FEM monitor $PM_{2.5}$ concentration (µg/m³)	$PM_{2.5}/PM_{10}$ Ratio FEM monitor concentrations	Average sensor PM_{10} concentration (µg/m³)	Averaged influence of RH on sensor measurements (µg/m³)
Effect of RH	Initial Testing Conditions	Setpoint	40 ± 5	20 ± 1	60 ± 5%		≤ 0.4 ± 0.1		
		Measured Value							
	High RH Conditions	Setpoint	85 ± 5	20 ± 1	60 ± 5%		≤ 0.4 ± 0.1		
		Measured Value							

1

Testing Report – PM_{10} Enhanced Testing Manufacturer & Air Sensor Name	Testing Organization Contact Email / Phone Number Date	Image of device during chamber evaluation

Effect of Temperature (T)

T Monitor	Manufacturer
	Model

			Average RH (%)	Average T (°C)	Average FEM monitor PM_{10} concentration of test aerosol (µg/m³)	Average FEM monitor $PM_{2.5}$ concentration (µg/m³)	$PM_{2.5}/PM_{10}$ Ratio FEM monitor concentrations	Average sensor PM_{10} concentration (µg/m³)	Averaged influence of T on sensor measurements (µg/m³)
Effect of T	Initial Testing Conditions	Setpoint	40 ± 5	20 ± 1	60 ± 5%		≤ 0.4 ± 0.1		
		Measured Value							
	High T Conditions	Setpoint	40 ± 5	40 ± 1	60 ± 5%		≤ 0.4 ± 0.1		
		Measured Value							

60-Day Low Concentration Drift

			Average RH (%)	Average T (°C)	Average FEM monitor PM_{10} concentration of test aerosol (µg/m³)	Average FEM monitor $PM_{2.5}$ concentration (µg/m³)	$PM_{2.5}/PM_{10}$ Ratio FEM monitor concentrations	Average sensor PM_{10} concentration (µg/m³)	Sensor drift after 60 days (µg/m³)
60-Day Low Concentration Drift	Day 1 Date	Setpoint	40 ± 5	20 ± 1	15 ± 10%		≤ 0.4 ± 0.1		
		Measured Value							
	Day 60 Date	Setpoint	40 ± 5	20 ± 1	15 ± 10%		≤ 0.4 ± 0.1		
		Measured Value							

60-Day Mid Concentration Drift

			Average RH (%)	Average T (°C)	Average FEM monitor PM_{10} concentration of test aerosol (µg/m³)	Average FEM monitor $PM_{2.5}$ concentration (µg/m³)	$PM_{2.5}/PM_{10}$ Ratio FEM monitor concentrations	Average sensor PM_{10} concentration (µg/m³)	Sensor drift after 60 days (µg/m³)
60-Day Mid Concentration Drift	Day 1 Date	Setpoint	40 ± 5	20 ± 1	60 ± 5%		≤ 0.4 ± 0.1		
		Measured Value							
	Day 60 Date	Setpoint	40 ± 5	20 ± 1	60 ± 5%		≤ 0.4 ± 0.1		
		Measured Value							

Testing Report – PM_{10} Enhanced Testing
Manufacturer & Air Sensor Name

Testing Organization
Contact Email / Phone Number
Date

Image of device during chamber evaluation

Accuracy at High Concentrations

		Average RH (%)	Average T (°C)	Average FEM monitor PM_{10} concentration of test aerosol (µg/m³)	Average FEM monitor $PM_{2.5}$ concentration (µg/m³)	$PM_{2.5}/PM_{10}$ ratio from FEM monitor concentrations	Average sensor PM_{10} concentration (µg/m³)	Test averaged difference between sensor and FEM PM_{10} concentrations (µg/m³)
Accuracy at High Concentrations	Setpoint	40 ± 5	20 ± 1	200 ± 5%		≤ 0.4 ± 0.1		
	Measured Value							
	Setpoint	40 ± 5	20 ± 1	300 ± 5%		≤ 0.4 ± 0.1		
	Measured Value							

3

Note: A fillable reporting template for enhanced testing is also available with this report. See accompanying PowerPoint File.

United States
Environmental Protection
Agency

PRESORTED STANDARD
POSTAGE & FEES PAID
EPA
PERMIT NO. G-35

Office of Research and Development (8101R)
Washington, DC 20460

Official Business
Penalty for Private Use
$300

Recycled/Recyclable Printed on paper that contains a minimum of
50% postconsumer fiber content processed chlorine free